# Market Access and Urban Growth in the Former Soviet Union[*]

Humoyun Abdumavlon[§]

**Abstract**

How does geopolitical fragmentation reshape urban growth? Using difference-in-differences on 1,396 cities over 1970–2021, I study the 1991 Soviet collapse. In post-Soviet Europe, annual population growth near new internal and old external borders fell by 0.4 and 0.6 percentage points relative to interior cities, implying populations 11–17% smaller after 30 years. Central Asian cities near previously sealed frontiers grew about one point faster. Manufacturing exposure amplifies internal-border losses; coastal access mitigates external-border losses. These patterns support a market-access interpretation in which fragmentation disrupts production links and market transition exposes locational disadvantages along existing frontiers.

*JEL classification:* F15, N94, O18, P25, R11, R12

*Keywords:* Market access; Urban growth; Borders; Former Soviet Union

[*]I am especially grateful to Adam Storeygard for his guidance and support. I also thank Omer Ali, Kristian Behrens, Daniel Berkowitz, Christa Court, Andreas Ferrara, Gordon Hanson, Yannis Ioannides, Lester Lusher, Enrico Spolaore, Gabriel Tourek, Randall Walsh, the NARSC Graduate Student Paper Competition Reading Committee, and participants at the 71st North American Meetings of the Regional Science Association International for their valuable discussions and suggestions.

[§]Department of Economics, University of Pittsburgh, United States of America. E-mail: abdumavlon@pitt.edu.

## 1 Introduction

Geopolitical fragmentation has created new states from Eritrea to South Sudan, yet credible causal evidence on its effects on urban growth in developing countries remains scarce. Identification is difficult because border realignments reflect local political and economic conditions and coincide with other disruptions.

This paper exploits the 1991 collapse of the Soviet Union as a natural experiment to causally estimate how border realignments affect the economic geography of cities. The administrative boundaries that separated the Soviet republics were drawn between the 1920s and 1940s, primarily based on ethno-linguistic criteria and centralized administrative convenience, rather than the economic potential or growth trajectories of individual cities (Kramer and Smetana, 2013). For seven decades, these lines imposed minimal friction on trade and migration (Eichengreen, 1993; Suesse, 2018). The dissolution was driven by aggregate, macro-level crises such as the collapse of global oil prices, union-wide fiscal failures, and political upheaval in Moscow, rather than the localized economic conditions of individual cities (Gaidar, 2010). The collapse caught even seasoned scholars off guard (Fukuyama, 1992; Smith, 1993; Kotkin, 2001). Because the location of these borders was historically predetermined and the timing of their enforcement was driven by macro-political forces, the resulting spatial shock to market access is plausibly exogenous to the pre-existing economic trajectories of individual cities.

Existing studies examine episodes in which market access moves in the same direction for all treated locations at a given moment, whether the shock is division or integration (Redding and Sturm, 2008; Ahlfeldt et al., 2015; Brülhart et al., 2018). German division lowered market access for border cities and reunification later raised it, but the two shocks arrive in sequence, so the aggregate environment differs across them. The Soviet collapse instead hardened some frontiers and opened others simultaneously, so the sign of the border shock varies across cities while union-wide transition shocks do not.

In post-Soviet Europe, newly enforced internal borders severed the dense industrial supply chains that Gosplan had constructed across republican boundaries, while the simultaneous collapse of COMECON[1] redirected former Eastern Bloc partners toward Western Europe. In Central Asia, the dissolution opened previously sealed frontiers with Afghanistan, China, and Iran, borders that had been closed or heavily militarized throughout the Cold War (Pomfret, 2010, 2021). While the Central Asian border sample is smaller and geographically concentrated, suggestive evidence indicates these cities grew rather than shrank. [2]

Figure 1 maps the sample, the two border types, each city's exposure, and the two regional groupings used throughout. I estimate difference-in-differences models comparing cities within 75 km of borders to interior cities, using population data for 1,221 cities in post-Soviet Europe and

[1]The Council for Mutual Economic Assistance (COMECON) was an economic organization comprising the Soviet Union and its socialist allies.

[2]Section 3 defines market access as the composite of demand access and supplier access, the two dimensions of proximity to markets that jointly determine equilibrium city population. I do not construct a structural index (e.g., Donaldson and Hornbeck, 2016). Such an index is impractical here because Gosplan's centralized material balance system violates the competitive market assumptions underlying standard gravity frameworks (Eaton and Kortum, 2002), and parameterizing the heterogeneous trade costs of newly hardened post-1991 borders is not feasible.

175 in Central Asia spanning 1970–2021. The long pre-period permits direct testing of parallel trends, and the three-decade post-collapse horizon captures persistent effects. I distinguish two border types: *internal borders*, the former inter-republican boundaries that became international after 1991, and *external borders*, the frontiers that were already international throughout the Soviet period. Newly international borders raised trade costs. At existing external frontiers, market transition could expose locational disadvantages previously buffered by administrative allocation, even as crossing became easier. The net effect depended on whether improved foreign access offset the loss of that insulation.

The main finding is that post-Soviet cities in Europe near internal and external borders see their populations decrease at annualized rates of about 0.4 and 0.6 percentage points (pp), respectively, although the external estimate is imprecise for cities not also near an internal frontier. These effects accumulate into relative contractions of roughly 11–17% over three decades. On the other hand, cities near Central Asia's external borders grew faster than those in the interior by about 1 pp per year, though the smaller sample warrants caution. Internal border effects in Central Asia are not distinguishable from zero and are significantly smaller than in Europe, consistent with the thinner cross-republican production linkages that ran through Central Asian cities before dissolution (Pomfret, 2021).

Three pieces of evidence are consistent with market access as the operative mechanism. First, internal border effects in Europe and the external gain in Central Asia attenuate with distance from the frontier, and Europe's external decline is concentrated among inland cities: it is significantly smaller, and not detected, among the 23 coastal cities, where a port offers an outlet the land frontier does not close. Second, I exploit sectoral heterogeneity in exposure to cross-border linkages: manufacturing-intensive cities, more dependent on the vertically integrated supply chains that spanned republican boundaries, experienced amplified losses where borders hardened in post-Soviet Europe, while cities whose production drew on locally sourced inputs show no border effect at all. Third, the results survive nearest-neighbour and exact matching on pre-1991 city size, industrial employment and composition, and defense sector presence, indicating that the estimates do not reflect pre-existing differences in observable characteristics.

This paper contributes to the new economic geography literature, which since Krugman (1991) has emphasized market access as a central determinant of the spatial distribution of economic activity (Helpman, 1998; Fujita et al., 2001; Redding and Venables, 2004). Empirical work has tested this prediction through border or administrative frictions (Behrens, 2024; Egger et al., 2022; Brülhart et al., 2018; Ahlfeldt et al., 2015; Redding and Sturm, 2008), transport infrastructure (Jedwab and Storeygard, 2022; Büchel and Kyburz, 2020; Baum-Snow et al., 2017; Storeygard, 2016; Faber, 2014), and trade policy (Dix-Carneiro and Kovak, 2017; Autor et al., 2013; Hanson, 2001). Closest to this paper, Suesse (2018) uses a gravity framework to show that Soviet disintegration severed inter-republican supply chains as nationalist mobilization advanced and documents the trade collapse at the republic level; I trace its spatial incidence across individual cities, so the population responses estimated here complement the trade-based results established at the aggregate.

This paper makes two contributions. First, it exploits a geopolitical rupture that simultaneously hardened some frontiers and opened others, so the sign of the market access change varies across cities within a single episode rather than across episodes, and it does so at a scale, nearly 1,400 cities across 15 countries, that permits tests of distance gradients and sectoral exposure. Second, it provides city-level evidence that the economic consequences of an existing frontier can change with the allocation regime. Cities along post-Soviet Europe's old external frontier declined relative to interior cities although these boundaries were not newly imposed and became relatively easier to cross. That market transition exposed the locational disadvantages of Soviet industrial siting is itself familiar (Hill and Gaddy, 2003); what is new is the evidence on international frontiers specifically, on the population trajectories of the cities beside them, and on the role of alternative access, since the penalty is significantly smaller where a port provides an outlet the land frontier does not close.

More broadly, the findings give city-level evidence on the costs of political fragmentation theorized by Alesina and Spolaore (1997, 2005), and on the persistence of city sizes after major disruptions (Miguel and Roland, 2011; Brakman et al., 2004; Davis and Weinstein, 2002): border realignments generate lasting spatial divergence where they durably rewire access to nearby markets.

## 2 Institutional Background

The dissolution of the Soviet Union in 1991 fragmented one of the most deeply integrated economies in the modern world. Since its founding in 1922, the USSR organized a vast and diverse population into 15 Soviet Socialist Republics through national-territorial delimitation (*razmezhevanie*). Administrative borders between republics imposed little friction on internal trade or migration (Eichengreen, 1993; Denisenko and Chudinovskikh, 2017). Republican agencies retained limited planning autonomy and could on occasion influence allocation within their territory, but their remit extended to the republic as a whole rather than to particular locations within it. Gosplan, the State Planning Commission, coordinated production through hierarchical negotiations, deliberately constructing vertically integrated industries that spanned republican boundaries. Djankov and Freund (2002) estimate a gravity equation for trade among Russian regions and FSU republics and find that Russian regions traded no more with one another than with other republics before the collapse. A pronounced domestic bias emerged afterward as inter-republican trade contracted, with Russian regions trading roughly 60% more among themselves than with the republics.

Spatial frictions in the Soviet economy were organized around individual cities rather than administrative lines: the *propiska* residency permit regime, introduced in 1932, restricted access to high-status cities such as Moscow, Saint Petersburg, and Tashkent rather than movement across republican borders (Buckley, 1995; Light, 2012; Gang and Stuart, 1999).

### *2.1 Soviet Europe*

The European republics contained the Union's most industrialized economies, far more densely integrated through planned supply chains than Central Asia (Pomfret, 2021), with factories often relying on single suppliers in other republics for critical inputs. Blanchard and Kremer (1997) demonstrate that when central planning's enforcement mechanism disappeared before market institutions emerged, bargaining broke down across these supply chains. Output fell farthest for goods with the most complex production processes, precisely because they depended on cross-republican coordination.

On the western frontier, Soviet external trade with Eastern Europe was organized through COMECON, which accounted for roughly half of Soviet foreign trade (Holzman, 1985).[3] Foreign trade as a whole was nonetheless small next to trade among the republics, which absorbed more than 80% of the exports of every republic except Russia in 1990 (Al-Atrash and Havrylyshyn, 1998; Djankov and Freund, 2002).

COMECON effectively dissolved on January 1, 1991, when members shifted to hard currency transactions. Intra-bloc trade dropped from 13 times gravity-predicted levels in 1990 to just 4 times by 1993 (Maurel and Cheikbossian, 1998). The western frontier also became harder to cross as the Central European states prepared for EU accession. Residents of Belarus, Moldova, Russia, and Ukraine could enter Poland without visas in the 1990s, but Poland introduced visas in October 2003, Romania required them of Moldovans on its accession in January 2007, and costlier Schengen visas followed in December 2007.

### *2.2 Central Asia*

Central Asian republics were integrated with Russia mainly through extractive north-south corridors for cotton, natural gas, and minerals, and lacked the dense inter-republican linkages characteristic of the European USSR (Pomfret, 2021), while over 80% of Soviet industrial output was concentrated in Russia and the western republics (IMF et al., 1991). All roads, railways, and pipelines led north to Russia.

Before 1991, Central Asian republics were nearly isolated from neighboring non-Soviet states. The Sino-Soviet border was closed in 1962 and remained heavily militarized through the Cold War (Radchenko, 2009), contact across the Iran-Turkmenistan border was channeled through Moscow, and the Afghanistan border became a war zone after the 1979 Soviet invasion; the first Central Asian rail link not routed through Russia opened only in 1990 (Pomfret, 2021). The dissolution opened all three frontiers simultaneously. Shuttle traders (*chelnoki*) built cross-border bazaar economies at newly accessible crossings (Spector, 2017), Iran and Turkmenistan began direct bilateral trade, and in 1992, when the Khorgos crossing on the Kazakhstan-China border reopened, approximately 700,000 Kazakhstani citizens crossed to China for "shop tourism" (Peyrouse, 2008).

[3]Trade with Türkiye and Iran remained minimal throughout the Cold War. The Armenian-Turkish border stayed closed after 1991 due to the Nagorno-Karabakh conflict. Türkiye and Azerbaijan share only a 17 km land border, offering little commercial potential in the early post-Soviet years, and Georgia's early-1990s wars further disrupted the frontier.

Appendix Figure A1 documents the initial reorientation: from 1991 the share of non-FSU trade conducted with former Eastern Bloc partners fell in the European republics and Russia, the share conducted with China and Afghanistan rose in the Central Asian republics, and interrepublic trade contracted to roughly one-fifth of its 1990 level by 1994, when the Soviet-format series ends.

The nature of the market access gained in Central Asia also differed qualitatively. Central Asian border cities gained access to lower-income neighbors: Afghanistan, Iran under sanctions, and China's least developed western provinces. As Kaminski and Mitra (2012) and Karrar (2019) document, the resulting cross-border trade was dominated by cheap consumer goods for resale in domestic bazaar economies, requiring little fixed investment and allowing gains to materialize rapidly through informal shuttle trade.

The Soviet collapse also triggered significant migration, particularly among ethnic minorities. Roughly a million ethnic Germans and Jews emigrated to Germany and Israel, while ethnic Russians returned to Russia from other republics (Becker et al., 2012; Heleniak, 2004). These flows were predominantly repatriations to ancestral homelands rather than localized sorting away from borders, and the largest emigrant populations were concentrated in major interior cities (Mukhina, 2007; Heleniak, 2004). Sections 3 and 5 examine whether such resettlement could account for the border effects.

## 3 Conceptual Framework

This section sets out the spatial equilibrium logic that organizes the analysis. I draw on the economic geography models of Krugman (1991), Helpman (1998), and Fujita et al. (2001), in the multi-region form used by Redding and Sturm (2008) and extended to intermediate inputs following Redding and Venables (2004). I use the framework to sign and interpret reduced-form estimates rather than to estimate structural parameters.

The framework describes adjustment under market allocation. Before 1991, Gosplan coordinated production and distribution, while the *propiska* system regulated mobility by city status. Administrative allocation could buffer locational disadvantages by sustaining links to distant suppliers and buyers. Market transition could therefore make access to nearby markets more consequential as some frontiers hardened and others opened. I assess pre-transition differences in growth empirically in Figures 2 and 3.

Workers are mobile and consume traded varieties and a local amenity in fixed supply. Firms produce differentiated varieties under monopolistic competition using labor and a composite intermediate, and both final and intermediate goods bear iceberg trade costs $\tau_{ic} > 1$. The price index of the intermediate bundle is the same trade-cost-discounted aggregate that governs the consumer cost of living, so a city's marginal cost is decreasing in its access to suppliers. Two distinct objects therefore summarize a city's position in this network. Supplier access is the trade-cost-discounted measure of varieties a city can obtain as inputs,

$$SA_c = \sum_i n_i \left(p_i \tau_{ic}\right)^{1-\sigma}, \tag{1}$$

and demand access is the trade-cost-discounted measure of expenditure its firms can reach,

$$DA_c = \sum_i (w_i L_i) P_i^{\sigma-1} \tau_{ci}^{1-\sigma}, \tag{2}$$

where $n_i$ is the mass of varieties produced in $i$ and $P_i$ its price index. Spatial equilibrium requires indirect utility to be equalized across populated cities, which yields equilibrium population as

$$L_c = \kappa H_c SA_c^{\gamma} DA_c^{\eta}, \qquad \gamma, \eta > 0, \tag{3}$$

where $H_c$ is the local amenity stock and $\gamma$ and $\eta$ depend on $\sigma$, the expenditure share on traded goods, and the intermediate share in production.

Supplier and demand access both depend on bilateral trade costs, so frontier hardening reduces both. Population responses cannot separately identify these margins; I call their composite *market access*. Supplier access is particularly relevant in the Soviet setting because input deliveries were difficult to replace (Blanchard and Kremer, 1997).

Within the market-allocation regime, the population response to a change in $\tau$ depends on the direction of that change and the share of a city's input purchases and output sales crossing the affected frontier. Where enforcement raised trade costs on a frontier carrying a large share of a city's linkages, equation 3 predicts decline; where a previously prohibitive frontier opened, it predicts growth. Opposite-signed regional effects therefore follow from one comparative static rather than separate theories, and the regional difference is a share parameter rather than a difference in institutions or planning regime. I test this with interactions for Central Asia.

At existing external frontiers, changes in crossing costs coincided with the transition away from administrative allocation. Planning could sustain links to distant suppliers and buyers, buffering the disadvantages of a truncated hinterland. Market transition could therefore increase the cost of peripheral location even where crossing became easier. Because equations 1 and 2 discount economic mass rather than realized trade, this argument does not require frontier cities to have conducted a disproportionate share of Soviet foreign trade. The net population response depends on whether improved foreign access offsets the loss of this insulation. In Europe, the contraction and westward reorientation of neighboring economies could limit that offset, while subsequent EU visa requirements could impede recovery (Section 2.1). Redding and Sturm (2008) likewise find smaller effects from German reunification than from division. This interpretation requires a change in the allocation regime as well as in border frictions. Equations 1 to 3 describe the market equilibrium, within which easier foreign access, holding other conditions fixed, raises both $SA_c$ and $DA_c$. The design does not identify the change in allocation regime separately from fragmentation itself.

In Central Asia the comparative static runs through the other margin. A lower share of production there ran through inputs sourced across republican boundaries, so enforcing internal frontiers severed fewer supplier links and the framework predicts a smaller internal effect, though individual republics departed from the regional average. What the opened external frontiers supplied was buyers, in the bazaar economies documented by Kaminski and Mitra (2012), and buyers enter $DA_c$ rather than $SA_c$. The European internal-border decline is therefore primarily

a loss of supplier access and the Central Asian external-border gain primarily an addition of demand access. What determines exposure is not sector as such but the share of a city's purchases and sales crossing the frontier whose $\tau$ changed. Manufacturing employment tracks that share in Europe but not in Central Asia, where manufacturing faced north toward Russia rather than toward the frontiers that opened, so low agricultural employment proxies instead for orientation toward cross-border commerce.

Beyond the sign, the framework predicts that effects should be localized, decaying with distance from the changed frontier, since cities farther away already faced high trade costs to locations across it. What decays is the hinterland a city forgoes, which depends on the geometry of the frontier and on city size, so the decline need not be strictly monotone in straight-line distance. The sea enters the same accounting. A coastline truncates a city's overland hinterland as a closed frontier does, but a port reaches distant foreign mass at low cost, so a land frontier accounts for a smaller share of a coastal city's access and should cost it less. Effects should also be larger for small cities, since own-city expenditure enters $DA_c$ at $\tau_{cc} = 1$ and is therefore a larger share of a large city's total access, so a given loss of external mass is proportionally smaller.

In the framework, population adjusts through internal migration: a loss of market access lowers real wages, inducing workers to leave until utility is equalized across cities. Observing migration does not, however, identify the shock that induced it. Ethnic resettlement, changes in mobility restrictions, and other local shocks correlated with border proximity could also generate differential population growth. The size, coastal-access, and sectoral comparisons in Section 5 assess whether the observed patterns support a market-access interpretation, but do not separately identify migration motives.

## 4 Data and Empirical Methodology

The empirical analysis draws on the following data sources:

1. Population census. The panel follows 1,396 former Soviet cities with over 5,000 residents in 1970 through 2021.[4] Population and coordinates come from Brinkhoff (2021), supplemented by Demoscope Weekly, the Russian Institute of Demography's database.[5] Observations cover the 1970, 1979, and 1989 Soviet censuses and three post-1991 periods, with dates varying by country (Table A1).

I use the annualized growth rate of population following Redding and Sturm (2008), as it allows comparing growth rates consistently across different census periods, adjusting for length variations in those periods.[6]

[4] The results are robust to using a threshold of 10,000.

[5] Available at citypopulation.de and demoscope.ru, respectively.

[6] It can be computed in city $c$ as

$$y_{ct} = \frac{ln(P_{ct}) - ln(P_{c,t-L_{ct}})}{L_{ct}}, \tag{4}$$

where $P_{ct}$ is the population statistic at time t, and $L_{ct}$ is the number of years between $t$ and the prior census. Log differences are symmetric, additive across periods, and nearly identical to the discrete compound annual growth rate at the typical growth rates in this sample (mean 0.68%).

For each city I also compute straight-line distances to the nearest internal border, the nearest external border, and the coast. External border distance is measured to the nearest non-FSU territory, which for all but three external border cities lies across a land frontier; Tallinn, Maardu, and Paldiski fall within 75 km of Finland across the Gulf of Finland. Coastal cities are those within 75 km of the open sea, excluding the landlocked Caspian.

2. Industry data from 1989. The paper uses total industrial employment, turnover, manufacturing output in the civilian industry, and the SIC code of each city's primary sub-industry from 1989, compiled by Kofanov and Mikhailova (2015) from Soviet statistical yearbooks. Matching their Cyrillic city names to the main sample, described in Appendix A1, recovers 1,225 of the 1,396 cities (1,076 of 1,221 in Europe and 149 of 175 in Central Asia); the unmatched are predominantly small places renamed after independence or without reported sectoral employment.

3. Defense industry data. Data on Soviet defense enterprises comes from Dexter and Rodionov (2024), who compiled a comprehensive database of military factories and research establishments across the USSR. I restrict the sample to enterprises whose military work ended between 1989 and 1991 and whose workforce size is documented, and construct for each matched city the count of such enterprises and the sum of their ordinal workforce sizes. Appendix A1 describes the filtering and name-matching.

Because sub-national GDP is unavailable across the FSU, Appendix C provides some corroborative results with nighttime lights from the Defense Meteorological Satellite Program (DMSP), 1992–2013, as a proxy for city-level economic activity (Henderson et al., 2012); Appendix A1 describes the cleaning. Finally, Appendix Tables A2 and A3 report descriptive statistics by region.

*Empirical Strategy*

The identifying variation is the differential change in population growth of border relative to interior cities after versus before the dissolution, pooled across post-Soviet countries that operated under the same centralized institutional framework before 1991. A more demanding specification replaces period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Eastern Europe, and Central Asia), restricting the comparison to border and interior cities *within the same sub-region and census period* and absorbing sub-region-specific macroeconomic trajectories.

What identification requires is weaker than the absence of friction. Whatever autonomy republican agencies exercised operated at the level of the republic and applied alike to its interior and its border cities, so it gives no reason for cities near administrative boundaries to follow different growth trajectories from interior cities in the same country.

Formally, the DiD approach does not require that border locations were randomly assigned but assumes that, absent the sudden dissolution, trends in the outcome variable would have been parallel between border and non-border cities (Roth et al., 2023; Meyer, 1995).[7]

The baseline specification is

[7]While spatial spillovers from border to interior cities technically violate SUTVA (Rubin, 1980), capturing this relative divergence is the goal of the DiD framework, consistent with spatial equilibrium models (Fujita et al., 2001; Crozet, 2004; Redding and Sturm, 2008).

$$y_{ct} = \alpha_c + d_t + \beta_1 \left(IB_c \times \text{Post}_t\right) + \beta_2 \left(EB_c \times \text{Post}_t\right) + \epsilon_{ct}, \tag{5}$$

where $IB_c$ and $EB_c$ equal one for cities within 75 km of an internal and an external frontier respectively, $\text{Post}_t$ equals one for census periods after 1991, $\alpha_c$ and $d_t$ are city and period fixed effects, and standard errors are clustered at the city level. The coefficients $\beta_1$ and $\beta_2$ give the change in annualized population growth for border relative to interior cities after dissolution. Columns (3)–(4) of Table 1 add a Central Asia indicator interacted with $\text{Post}_t$ and with each treatment, so that $\beta_1$ and $\beta_2$ report effects for post-Soviet Europe and the triple interactions report the Central Asian differences.

To validate the parallel trends assumption and examine dynamic effects over time, I employ an event-study framework:

$$y_{ct} = \alpha_c + d_t + \sum_{k=T_0}^{-2} \delta_{1,k}\left[IB_c \times D_k\right] + \sum_{k=0}^{T_1} \delta_{1,k}\left[IB_c \times D_k\right] + \sum_{k=T_0}^{-2} \delta_{2,k}\left[EB_c \times D_k\right] + \sum_{k=0}^{T_1} \delta_{2,k}\left[EB_c \times D_k\right] + \epsilon_{ct} \tag{6}$$

where $y_{ct}$ is the annualized growth rate of population in city $c$ in census period $t$,[8] $IB_c$ and $EB_c$ are defined as in equation 5, and $D_k$ is an indicator equal to one when the observation falls in event-time period $k$ relative to the Soviet dissolution (with $k = 0$ denoting the first post-collapse census period).

The window is fixed at $k \in [-2, +2]$ with $k = -1$ excluded, so coefficients at $k < -1$ capture pre-trend dynamics and those at $k \geq 0$ the evolution of the treatment. Because post-Soviet countries conducted censuses at different dates, the time index denotes relative census periods rather than calendar years. I do not use country-by-period fixed effects. The successor states operated as a single planned economy before 1991, and in the smaller republics nearly every city lies near a frontier, so such fixed effects would absorb much of the treatment there; the estimates therefore include the national component of each state's exposure to the new borders, which the design cannot separate from other national shocks in those states.

## 5 Results

*DiD Estimates*

Table 1 begins the analysis with the pooled sample of all 1,396 cities. Columns (1)–(2) impose a common border effect across regions. Column (1) estimates annualized declines of 0.21 pp for cities near internal borders and 0.563 pp for cities near external borders. Because smaller republics such as the Baltic and Caucasian states have mechanically higher shares of border cities, common period fixed effects could conflate national trajectories with border effects. Column (2) therefore replaces them with sub-region-by-period fixed effects. The external border effect is nearly unchanged at $-0.540$ pp, significant at the 1% level, while the internal border effect falls to $-0.159$ pp and loses significance. Part of this attenuation reflects the common coefficient,

[8] It covers five intercensal periods – 1970-1979, 1979-1989, 1989-1999, 1999-2009, and 2009-2019 – with variations after the collapse depending on specific country census years.

which averages a negative European effect against a Central Asian effect indistinguishable from zero (Columns 3–4); the rest reflects the sub-regional trajectories these fixed effects absorb.

Columns (3)–(4) add the Central Asia interactions. The European internal and external border effects are −0.397 and −0.591 pp under period fixed effects and −0.241 and −0.663 pp under sub-region-by-period fixed effects, and a joint test rejects that both triple interactions equal zero ($p = 0.006$ and $p = 0.002$). The Central Asia main effect is 0.079 pp ($p = 0.807$), so the regional contrast at the borders does not reflect a detectable difference in interior-city trajectories.

The implied Central Asian effects reported at the foot of the table show where that difference lies. Internal border effects are 0.411 and 0.471 pp and are not distinguishable from zero ($p = 0.256$ and $p = 0.201$), consistent with the limited inter-republican linkages that existed before dissolution (Pomfret, 2021). External border cities grew 0.781 pp ($p = 0.074$) and 0.969 pp ($p = 0.031$) faster than interior cities, implying a relative cumulative population increase of roughly one-third over three decades. This sample is small and geographically concentrated: 12 of the 15 external border cities lie along the Afghanistan frontier, with two near the Iran–Turkmenistan border and one near the Kazakhstan–China border. The consistency of positive point estimates across these distinct segments is reassuring, but the estimates warrant caution. Columns (5)–(6) estimate the regions separately. Because Central Asia is one of the four sub-regions, Column (4) already estimates region-specific effects, and its implied Central Asian effects reproduce Column (6) exactly.

*Event-Study Estimates*

Figure 2 indexes total border and interior city population to 1970 and plots their difference, separately by border type and region. In all four panels border and interior cities move in parallel before 1991. Afterwards, European border cities decline sharply and persistently relative to the interior, more so near external borders, while in Central Asia border cities grow faster in relative terms, most visibly near external borders.

Figure 3 presents event-study estimates from equation 6, with $k = -1$ as the omitted reference period. In Europe, the pre-collapse coefficient at $k = -2$ is close to zero for both border types, supporting the parallel trends assumption. In Central Asia, internal border pre-trends are similarly flat, while external border cities show a marginally significant negative coefficient at $k = -2$, which may reflect noise in a small sample but warrants caution in interpreting the Central Asian external border results.

In Europe, both border types show sharp and persistent declines after dissolution, with effects peaking at $k = 1$ and remaining strongly negative through $k = 2$. For external borders, the persistence through $k = 2$ is consistent with the westward reorientation of the economies across the frontier. For internal borders, the delayed peak is consistent with Blanchard and Kremer (1997)'s account of cascading supply chain failures. The persistence of effects three decades after dissolution indicates lasting spatial divergence rather than temporary disruption.

In Central Asia, the pattern reverses. Internal border effects remain small and statistically insignificant across all post-dissolution periods, consistent with limited inter-republican trade to

disrupt (Pomfret, 2010). External border cities show a gradual emergence of positive effects. The coefficients at $k = 0$ and $k = 1$ are positive but imprecisely estimated, consistent with the initially informal and uneven nature of cross-border commerce as shuttle traders began exploiting newly opened frontiers (Kaminski and Mitra, 2012). By $k = 2$, the effect becomes significant at the 5% level.

Figure A2 confirms that the event-study patterns are robust to replacing period fixed effects with sub-region-by-period fixed effects in Europe and to excluding Turkmenistan in Central Asia. The external border effects in Europe are, if anything, larger and more precisely estimated under this more demanding specification.

*Robustness*

First, 80 European cities lie within 75 km of both frontiers (56% of external and 23% of internal border cities), and the baseline switches both indicators on for them. Table A4 therefore partitions cities into internal-only, external-only, dual-exposure, and interior categories. In Europe, internal-only cities decline by 0.358 pp ($p = 0.001$) and dual-exposure cities by 1.099 pp ($p < 0.01$). The 62 external-only cities decline by 0.441 pp, but with a standard error of 0.291 this estimate can reject neither zero nor the baseline of 0.591 pp. The baseline external coefficient therefore draws much of its precision from dual-exposure cities, where external exposure adds −0.740 pp relative to internal-only cities ($p = 0.007$), or −0.875 pp ($p = 0.001$) under sub-region-by-period fixed effects. The dual-exposure decline is larger in magnitude than the sum of the single-exposure declines, but the test does not reject additivity ($p = 0.456$). Under sub-region-by-period fixed effects neither single-exposure coefficient is distinguishable from zero (−0.172 and −0.437 pp), so within-sub-region identification of both border effects rests substantially on doubly exposed cities.

In Central Asia (Columns 3–4), only 5 cities lie near both frontiers, so the dual-exposure coefficient supports little inference. The external-only estimate is 1.356 pp ($p < 0.05$), and 1.192 pp ($p < 0.10$) excluding Turkmenistan, both above the pooled 0.969 pp in Table 1. The internal-only estimate is 0.552 pp and insignificant, but 0.778 pp ($p < 0.05$) excluding Turkmenistan, so the Central Asian internal estimate is sensitive to sample composition.

Table A5 splits the sample at the median 1970 city population, and the European effects are carried almost entirely by small cities (column 3). Below the median, internal border cities decline by 0.447 pp and external border cities by 0.587 pp, both significant at the 5% level. Above the median the estimates fall to negative 0.130 and 0.200 pp and neither is distinguishable from zero. The Central Asian external border gain follows the same pattern, with small cities growing 1.402 pp faster.

The larger point estimates for small cities are consistent with Section 3 and the pattern reported by Redding and Sturm (2008) for German division. This pattern is less readily explained by an account centered on departures from large urban centers, but city size alone cannot distinguish economic migration from ethnic resettlement or changes in mobility restrictions.

If market transition raised the cost of peripheral location, the external border penalty should be smaller where maritime access offers an alternative route to suppliers and customers. Table A6

uses each city's distance to the open-sea coastline. Allowing the 164 cities within 75 km of the coast their own post-1991 trajectory leaves the European estimates at −0.397 and −0.592 pp (Column 1), and dropping them yields −0.368 and −0.784 pp (Column 3). The estimates are essentially unchanged under sub-region-by-period fixed effects and under alternative coastal definitions (Columns 2 and 5–7). Cities on the coast itself did not outgrow the interior after 1991 (−0.445 pp at 25 km, Column 5), plausibly because many Soviet ports were naval, fishing, or northern settlements. Column 4 lets the border effects differ by sea access. Inland external border cities decline by 0.782 pp, while the 23 coastal ones show no detectable decline (0.377 pp; $p = 0.007$ for equality). Internal border effects do not differ by coastal access (−0.383 and −0.366 pp, $p = 0.951$), consistent with a shock that runs through cross-republican supply links a port does not readily replace.

Table A7 reports additional sensitivity checks. The results are robust to excluding large cities subject to *propiska* restrictions, which also addresses ethnic composition since these cities hosted disproportionately diverse populations and experienced the highest outmigration rates after 1991 (Heleniak, 2004) (Column 1); to dropping the turbulent 1989–1999 intercensal period, after which the European external border effect is -0.648 pp, so the external effect is not a product of the disruption of the early 1990s (Column 2). Column (3), which drops dual-exposure cities, reproduces the single-exposure estimates in Table A4.

Column (4) excludes conflict-affected border pairs (e.g., Armenia–Azerbaijan and Georgia–Russia). Removing conflict zones *strengthens* the European internal border effect to -0.690 pp ($p < 0.01$), showing that the estimate does not depend on including those conflict-affected pairs. The external border effect remains marginally significant (−0.412 pp, $p = 0.071$).

Column (5) applies a region-specific restriction addressing how the regions are classified. In Europe it drops the 193 Russian cities in the Siberian, Far Eastern, and Urals federal districts; the internal border effect strengthens to −0.532 pp and the external effect holds at -0.534 pp, both significant at the 1% level, so the European estimates do not depend on grouping Asian Russia with post-Soviet Europe. In Central Asia the column drops the ten cities on the Kazakhstan–Russia frontier, the one Central Asian internal border that severed a dense industrial corridor. The internal estimate rises from 0.471 to 0.569 pp and remains insignificant.

Column (6) restricts the control group to cities within 750 km of any border. In Europe the internal negative effect strengthens to −0.529 pp while the external effect attenuates to -0.392 pp but remains significant at the 5% level, consistent with a peripherality component in the external estimate that the restriction partly removes (see the distance results below). In Central Asia neither effect moves much.

Allowing for spatial correlation across nearby cities, Conley standard errors at cutoffs from 25 to 200 km preserve all significance levels in both regions (Table A8; Conley, 1999; Colella et al., 2023). Figure A3 presents leave-one-out analyses. In Europe, both border effects remain negative across all ten country exclusions. The external estimate is smallest and least precise when Russia is dropped, which removes most of the interior control group. In Central Asia, the external border coefficient remains positive across all five exclusions, but both effects are

sensitive to dropping Kazakhstan, which contributes the largest share of interior control cities and leaves too few for precise estimation.

Appendix Table A9 replaces the binary treatment with 25 km distance bins, following Redding and Sturm (2008). In Europe, internal border effects decay with distance, from a decline of 0.486 pp at 0–25 km (significant at 1%) to 0.363 pp at 25–50 km (5%), and are insignificant beyond 50 km. External border effects are largest at 0–25 km (−1.055 pp) but non-monotone beyond it: the 25–50 km coefficient is insignificant, while the 75–100 km coefficient (−0.835 pp) is nearly as large as the first. This irregular gradient is consistent with a broader peripherality component if market transition increased the importance of access to nearby markets. I therefore rely on internal borders for the distance-gradient evidence of a localized frontier effect. Under sub-region-by-period fixed effects (Column 2) the external gradient strengthens while internal border effects attenuate. In Central Asia, cities within 25 km of external borders grew 1.438 pp faster (significant at 5%), with positive but imprecise coefficients farther out, consistent with the localized shuttle trade documented by Kaminski and Mitra (2012).

Table A10 redefines the binary treatment at thresholds from 25 to 125 km. The European internal border effect attenuates monotonically, from -0.537 pp at 25 km to -0.178 pp at 125 km, as a frontier-specific shock implies, whereas the external effect increases from -1.055 to -0.595 pp and then declines to -0.833 pp, with the standard error more than halving, consistent with the same peripherality component. In Central Asia the external gain is significant at the 5% level at every threshold up to 100 km (15 treated cities at 75 km, 19 at 100 km), while the internal estimate is insignificant throughout.

The wider rings also address the Soviet 101st-kilometre restriction, which limited residence around major cities rather than near frontiers, and the results are robust to excluding the cities it surrounded (Table A7, Column 1).

Appendix B reports border-pair estimates: most European internal pairs are negative and all Central Asian external pairs positive, though pairs with few cities are imprecise. Appendix C reports nighttime lights from 1992, which reproduce the cross-regional contrast at external borders in the first post-collapse decade but otherwise corroborate the population results only partially.

*Sectoral Heterogeneity*

I test the sectoral prediction of Section 3 using 1989 industrial employment by sector (Kofanov and Mikhailova, 2015), decomposing the European border effects by manufacturing quartile and splitting Central Asian cities at the within-region median of agricultural employment.[9] Appendix Table A12 reports median splits for all remaining sector-by-border-type combinations.

In Europe (Table 2, Columns 1–2), manufacturing shares are compressed (mean 0.834, median 0.925), so a linear interaction is hard to interpret and each quartile receives its own treatment effect. Lowest-quartile cities near internal borders show no significant decline (+0.305, $p = 0.216$), while all three upper quartiles, each with manufacturing shares above 0.75, decline

[9]Summary statistics for the matched sample, including the distribution of cities across heterogeneity bins by border type, are reported in Appendix Table A11.

by 0.5 to 0.8 pp, individually and jointly significant ($p = 0.007$). The effects do not rise from Q2 to Q4, which suggests a threshold rather than a dose-response.

At external borders Q4 cities decline more than Q1 ($-0.923$ vs. $-0.255$ pp), but the quartile interactions are imprecise (joint test $p = 0.691$). Less sectoral variation is expected there, since a truncated hinterland penalizes a city whatever it produces, whereas a severed supply chain penalizes the sectors that relied on it.

Columns (3)–(4) turn to Central Asia. The manufacturing split is uninformative there, since only 2 of the 11 treated external border cities fall in the high group ($p = 0.997$; Appendix Table A12), whereas the agriculture split gives cells of 8 high and 3 low. Low-agriculture external border cities grew 3.27 pp faster ($p = 0.006$), high-agriculture ones showed no significant gain (0.647, $p = 0.324$), and the difference is significant ($p = 0.047$). With three treated cities in the low group, this is indicative rather than precise. Internal border effects in Central Asia are insignificant across all sectors.

Table A12 reports the remaining sectors. Extraction is the clearest case of a low share. Mining cities do buy inputs, including equipment, explosives, fuel and timber, but they buy them intermittently from a small number of suppliers, whereas an assembly plant needs continuous delivery of hundreds of specific components and halts when any one fails to arrive. This is the distinction Blanchard and Kremer (1997) identify, with output falling farthest for goods with the most complex production processes. Consistent with it, European cities with any mining employment show no decline at internal borders ($+0.059$, $p = 0.762$) while cities without mining decline by 0.578 pp ($p < 0.001$), and the difference is significant at $p = 0.011$. The same ordering appears at external borders ($-0.170$ against $-0.915$ pp, $p = 0.090$).

Agriculture behaves differently. European cities with agricultural employment decline by 0.307 pp against 0.446 pp for those without, a difference not distinguishable from zero ($p = 0.570$), so there is no evidence that agriculture insulates European cities as mining does. This is plausible because Soviet agro-processing was embedded in cross-republican supply chains for machinery, fertiliser and onward processing. In Central Asia the sector matters instead through the demand margin (Table 2, Column 4): what separates cities is not whether a sector is primary or industrial but which margin of market access the 1991 rupture moved for it.

Two further sectors test the prediction where it should fail to bite. Utilities, almost entirely electric power, district heating, gas distribution and water supply, are produced and consumed locally with locally sourced inputs. European cities with utility employment show no detectable border decline at either frontier ($-0.131$ and $-0.018$ pp), while cities without them decline by 0.585 and 1.088 pp, with equality rejected at $p = 0.090$ and $p = 0.035$. Industrial repair and maintenance has local inputs but no independent exposure to cross-border linkages in either direction, and here the two groups are indistinguishable ($-0.413$ against $-0.368$ pp, $p = 0.854$). Across the four sectors, exposure thus rises with the density of cross-border input linkages. These are the only sectors the 1989 data permit, since construction, retail and public administration are empty or near-empty.

*Alternative Accounts*

I address compositional alternatives by selecting comparison samples through propensity-score matching. Matching is conducted separately by region and border type on the city cross-section. Probit specifications use 1970 population, 1989 industrial employment, turnover, or manufacturing output separately, or defence enterprise count and workforce jointly. Each treated city is matched, with replacement and under common support, to up to two cities unexposed to that border type, chosen by proximity in propensity score. I then re-estimate the baseline difference-in-differences on the selected cities' full population panels. Each selected city enters once per period, without weighting controls by their frequency of selection. An exact-match restriction retains cities in 1989 SIC categories containing both treated and control observations. Additional European specifications match jointly on all six covariates, with variants restricting eligible controls to within 500 or 1,000 km of each treated city.

Table 3 reports the European estimates for the six matching exercises described above.[10] Internal border declines range from 0.36 to 0.58 pp, bracketing the baseline estimate of 0.40 pp, and external border declines from 0.53 to 0.90 pp, all statistically significant, indicating that the estimates are robust to these matching restrictions.[11]

Column (6) addresses military divestment specifically. The matched sample is smaller (267 cities for internal borders, 100 for external) but the border effects persist at similar magnitudes: −0.46 pp for internal borders and −0.83 pp for external borders, both statistically significant. The Central Asian matched samples (Appendix Table A14) contain only 5 to 25 cities and support no systematic inference.

Table A15 varies the matching design itself. Column (1) conditions jointly on all six covariates, yielding −0.505 pp for internal borders on the 1,076 cities with complete covariate data, and Column (4) gives −0.802 pp for external borders. Balance in these matched samples is close: mean absolute standardized bias falls from 11.5 to 4.1 percent, and a probit of treatment on all six covariates in the matched sample has a pseudo $R^2$ of 0.001 and cannot reject that the two groups are identical ($\chi^2(6) = 1.17$, $p = 0.978$).

Figure A4 maps the matches. Selected controls are geographically dispersed, but proximity in propensity score does not ensure geographic proximity: the median pair distance is 1,326 km. Following Redding and Sturm (2008), who impose a geographic band on their control group, Columns (2)–(3) and (5)–(6) of Table A15 require each control to lie within 500 or 1,000 km of its treated city. Every treated city retains two matched controls even at 500 km, where the median pair distance falls to 347 km, and the estimates are stable: −0.392 and −0.600 pp for internal borders and −0.526 and −0.699 pp for external borders, against baseline estimates of −0.402 and −0.595 pp.

Finally, Soviet Science Cities and other closed settlements lay overwhelmingly in the interior, and Schweiger et al. (2022) show that Science Cities outperformed comparable localities after 1991, so as controls they could inflate the estimated border penalties. Excluding sample cities

[10] Appendix Table A13 reports covariate balance.

[11] Synthetic control, the natural alternative in border settings, is built for a few treated units observed over a long pre-treatment series, whereas there are 353 treated cities along internal frontiers in post-Soviet Europe and three pre-1991 census observations, and the object of interest is an average effect across a large treated group.

within 3 to 20 km of the 91 locatable Science Cities, and 20 further closed or restricted-access cities elsewhere in the union, leaves the European internal effect between −0.361 and −0.395 pp and the external effect between −0.592 and −0.609 pp, and the Central Asian external gain at 0.833 pp, significant at the 10% level (Table A16).

## 6 Conclusion

This paper examines how geopolitical fragmentation affects urban growth, exploiting the Soviet collapse as a natural experiment across nearly 1,400 cities and a five-decade horizon. In post-Soviet Europe, cities near the new internal borders and the old external frontier experienced persistent population declines relative to interior cities; in Central Asia, cities near previously sealed external borders grew faster. The distance, sectoral, size, and coastal patterns are consistent with market access as an important channel.

The spatial consequences of state dissolution extend beyond newly created boundaries: changes in allocation institutions can alter the economic importance of existing frontiers, exposing disadvantages previously buffered by administrative coordination. Border changes are not mechanically harmful or beneficial for nearby places: where a boundary cuts through integrated production networks, nearby cities experience persistent relative decline, and where it unlocks access to previously closed markets they can experience sustained relative growth. Because the effects concentrate among smaller cities, geopolitical fragmentation widens spatial inequality, disproportionately penalizing cities least equipped to absorb the loss of cross-border market access. As movements for regional autonomy continue globally, this asymmetry deserves attention from policymakers weighing the spatial economic costs of fragmentation.

## Figures & Tables




Figure 1: City Types and Regional Classification in the Former Soviet Union

*Notes:* The map shows all 1,396 cities in the sample. Cities are classified into four mutually exclusive categories by border exposure: within 75 km of a former inter-republican (internal) border only, within 75 km of an external border only, within 75 km of both, and interior cities beyond 75 km of any border. Blue lines mark internal borders, the boundaries between post-Soviet states that were administrative before 1991; black lines mark external borders, the frontiers that were already international throughout the Soviet period. Fills distinguish the two regional groupings used in the analysis: post-Soviet Europe comprises Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Russia, and Ukraine; Central Asia comprises Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan, following Council of the European Union (2009). Russia is classified with post-Soviet Europe despite spanning both continents: the grouping reflects the density of cross-republican production linkages rather than physical geography, and Russian cities in Asia are treated on the same basis as all others, facing internal frontiers with Kazakhstan and external frontiers with China and Mongolia. A small number of far-eastern Russian cities in the estimation sample lie outside the plotted extent. The 75 km cut-off follows Redding and Sturm (2008).

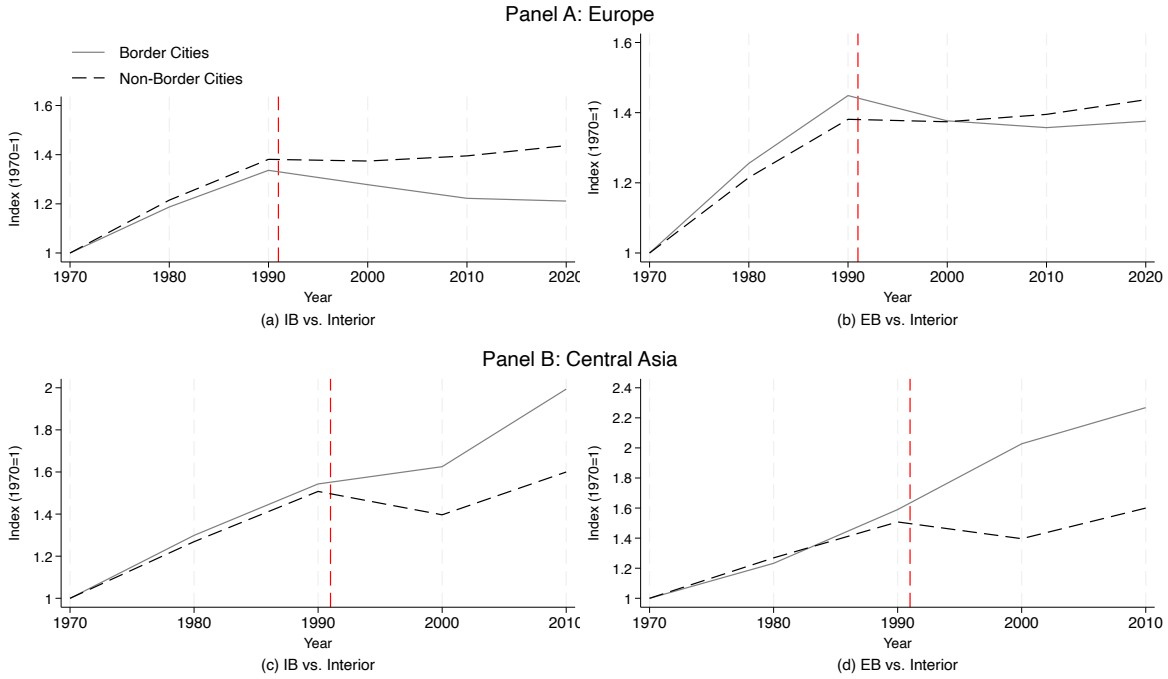



Figure 2: Indices of Border and Non-Border City Population

*Notes: Panel A includes all post-Soviet countries geographically situated in Greater Europe: Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Russia, and Ukraine. Moldova is excluded due to an unbalanced panel. The Central Asia panel includes all five post-Soviet Central Asian countries, but the balanced index shown here includes only two post-collapse periods, as Turkmenistan and Uzbekistan conducted only two censuses after the Soviet dissolution.*

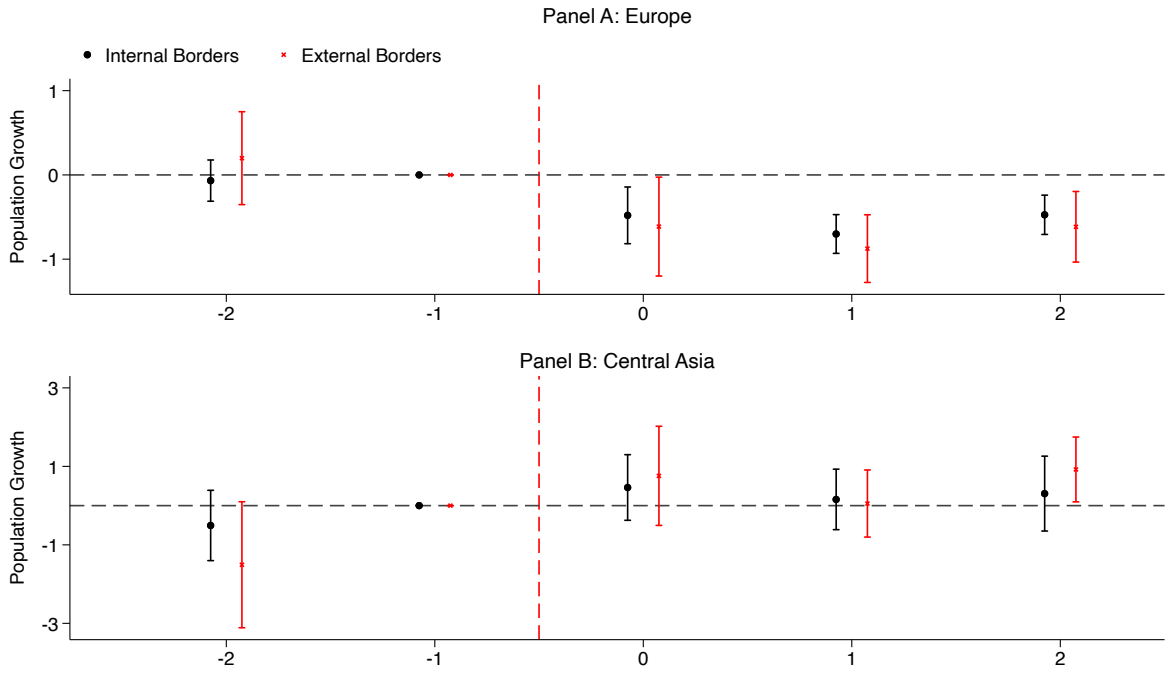



Figure 3: Event-Study Estimates: Baseline

*Notes: Coefficient plots from event-study regressions of annualized population growth on border proximity interacted with census-period indicators (equation 6). Black circles denote internal borders; red crosses denote external borders. Panel A shows Europe; Panel B shows Central Asia. The omitted period is $k = -1$ (1979–1989). All specifications include city and period fixed effects. Control group: interior cities beyond 75 km from any border. Standard errors clustered at the city level; bars show 95% confidence intervals.*

**Table 1: Urban Growth and Market Access: Main Results**

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| IB × Division | −0.210** | −0.159 | −0.397*** | −0.241** | −0.402*** | 0.471 |
| | (0.105) | (0.112) | (0.111) | (0.114) | (0.111) | (0.370) |
| EB × Division | −0.563*** | −0.540*** | −0.591*** | −0.663*** | −0.595*** | 0.969** |
| | (0.190) | (0.184) | (0.200) | (0.196) | (0.200) | (0.451) |
| CA × Division | | | 0.079 | | | |
| | | | (0.323) | | | |
| IB × Division × CA | | | 0.808** | 0.712* | | |
| | | | (0.378) | (0.386) | | |
| EB × Division × CA | | | 1.373*** | 1.633*** | | |
| | | | (0.480) | (0.490) | | |
| Implied IB effect, CA | | | 0.411 | 0.471 | | |
| *p*-value | | | 0.256 | 0.201 | | |
| Implied EB effect, CA | | | 0.781 | 0.969 | | |
| *p*-value | | | 0.074 | 0.031 | | |
| *p*: CA interactions jointly zero | | | 0.006 | 0.002 | | |
| Outcome mean (%) | 0.676 | 0.676 | 0.676 | 0.676 | 0.583 | 1.369 |
| Sample | All | All | All | All | Europe | CA |
| Cities | 1,396 | 1,396 | 1,396 | 1,396 | 1,221 | 175 |
| Observations | 6,809 | 6,809 | 6,809 | 6,809 | 6,004 | 805 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | | ✓ | | ✓ | ✓ |
| Sub-Region-by-Period FE | | ✓ | | ✓ | | |

*Notes:* The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders, respectively; "CA" denotes Central Asia (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan). "Division" equals one for post-1991 periods. Columns (1)–(4) pool all 1,396 cities. Column (1) includes city and period fixed effects. Column (2) replaces period fixed effects with sub-region-by-period fixed effects, where sub-regions are the Baltics, the Caucasus, Slavic Europe plus Moldova, and Central Asia. Columns (3)–(4) add interactions between the Central Asia indicator and each treatment, so the IB and EB coefficients report effects for post-Soviet Europe and the triple interactions report Central Asian differences relative to Europe; the Central Asia main effect is collinear with the sub-region-by-period fixed effects and is therefore omitted in Column (4). Implied Central Asian effects are the corresponding coefficient sums, with *p*-values from Wald tests of the linear combination, and the final row reports a joint test that both Central Asian interactions equal zero. Columns (5)–(6) estimate the baseline specification on each region separately. Because Central Asia constitutes one sub-region, Column (4) is numerically equivalent to estimating the two regions separately with sub-region-by-period fixed effects: its IB and EB coefficients and its implied Central Asian effects reproduce those separate estimates exactly. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table 2: Urban Growth and Sectoral Composition

| | Dependent Variable: Population Growth | | | |
|---|---|---|---|---|
| | Europe: Manufacturing Quartiles | | Central Asia: Agriculture Median Split | |
| | (1) Internal | (2) External | (3) Internal | (4) External |
| Border × Division (Q1, ref.) | 0.305 | −0.255 | | |
| | (0.247) | (0.520) | | |
| Border × Division × Q2 | −1.102*** | −0.313 | | |
| | (0.335) | (0.627) | | |
| Border × Division × Q3 | −0.912*** | −0.659 | | |
| | (0.334) | (0.619) | | |
| Border × Div. × Q4 (highest) | −0.813** | −0.668 | | |
| | (0.368) | (0.738) | | |
| Q2 × Division | 0.460** | 0.460** | | |
| | (0.184) | (0.184) | | |
| Q3 × Division | 0.491*** | 0.491*** | | |
| | (0.183) | (0.183) | | |
| Q4 × Division | 0.474** | 0.474** | | |
| | (0.228) | (0.228) | | |
| Border × Div. × High Agric. | | | 0.554 | 0.647 |
| | | | (0.584) | (0.651) |
| Border × Div.× Low Agric. | | | 0.340 | 3.268*** |
| | | | (0.642) | (1.135) |
| High Agric. × Division | | | 0.273 | 0.259 |
| | | | (0.782) | (0.789) |
| $p$: joint Q2=Q3=Q4=0 | 0.007 | 0.691 | | |
| $p$: Q4 vs Q1 | 0.028 | 0.366 | | |
| $p$: High = Low | | | 0.805 | 0.047 |
| Outcome mean (%) | 0.527 | 0.578 | 1.362 | 1.328 |
| Cities | 1,021 | 848 | 141 | 61 |
| Observations | 5,033 | 4,206 | 652 | 284 |
| City FE | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ |

*Notes:* The dependent variable is annualized city population growth (%). The matched industry sample contains 1,076 European cities. Columns (1)–(2) exclude cities near external borders only (*ext_only*=1), so that the internal border effect is identified against interior cities not exposed to external border shocks. Columns (3)–(4) similarly exclude cities near internal borders only. This accounts for the difference between the 1,076 matched cities and the column-specific sample sizes. Columns (1)–(2) decompose the European border effect by 1989 manufacturing employment quartiles, where Q1 (lowest manufacturing share) is the reference category. The quartile cutpoints are computed over all European cities with sectoral data. Column (1) estimates the internal border effect; Column (2) estimates the external border effect. Columns (3)–(4) split Central Asian border cities at the within-region median of 1989 agricultural employment share; "High" denotes cities above the median and "Low" denotes cities at or below the median. Because the median agricultural share is close to zero, the split effectively compares cities with any substantial agricultural employment to those without. "Border" refers to cities within 75 km of an internal border in Columns (1) and (3) and an external border in Columns (2) and (4). "Division" equals one for post-1991 periods. Columns (1) and (2) exclude cities near external and internal borders only, respectively. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Table 3: Robustness to Matching on Pre-1991 Characteristics (Europe)**

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A: Internal Borders** | | | | | | |
| IB × Division | −0.576*** | −0.361** | −0.571*** | −0.443*** | −0.381*** | −0.463** |
| | (0.128) | (0.143) | (0.129) | (0.136) | (0.125) | (0.199) |
| Matching on | 1970 Pop. | 1989 Ind. Empl. | 1989 Ind. Turn. | 1989 Manuf. | 1989 SIC | Military |
| Outcome mean (%) | 0.568 | 0.542 | 0.457 | 0.471 | 0.512 | 0.537 |
| Cities | 737 | 655 | 651 | 657 | 945 | 267 |
| Observations | 3,601 | 3,207 | 3,187 | 3,217 | 4,654 | 1,309 |
| Adj. $R^2$ | 0.427 | 0.432 | 0.433 | 0.436 | 0.454 | 0.474 |
| **Panel B: External Borders** | | | | | | |
| EB × Division | −0.528** | −0.690*** | −0.901*** | −0.623** | −0.626*** | −0.826** |
| | (0.247) | (0.258) | (0.248) | (0.252) | (0.237) | (0.409) |
| Matching on | 1970 Pop. | 1989 Ind. Empl. | 1989 Ind. Turn. | 1989 Manuf. | 1989 SIC | Military |
| Outcome mean (%) | 0.747 | 0.637 | 0.574 | 0.564 | 0.566 | 0.745 |
| Cities | 310 | 254 | 278 | 275 | 752 | 100 |
| Observations | 1,512 | 1,243 | 1,360 | 1,346 | 3,726 | 490 |
| Adj. $R^2$ | 0.389 | 0.365 | 0.379 | 0.397 | 0.439 | 0.309 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* Each column re-estimates the baseline specification on a matched subsample of treated border cities and their nearest neighbors. "Cities" in the matching tables reports the number of unique cities in the matched sample; "Observations" equals cities multiplied by census periods. Columns (1)–(4) and (6) use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section: each treated (border) city is paired with its two closest interior cities based on probit-estimated propensity scores for the indicated covariate. Because matching is with replacement, a control city may serve as a neighbor for multiple treated cities but enters the estimation sample once. Column (5) uses exact matching on each city's primary SIC code, restricting the sample to border and interior cities that shared the same dominant industrial classification in 1989; the larger sample reflects the retention of all cities within shared SIC strata rather than a fixed number of neighbors per treated city. "Military" matches jointly on the count and total workforce of defense enterprises that closed between 1989 and 1991. The dependent variable is annualized city population growth (%). "Division" equals one for post-1991 periods. All specifications include city and period fixed effects. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## References


Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., and Wolf, N. (2015). The economics of density: Evidence from the Berlin Wall. *Econometrica*, 83(6):2127–2189.

Al-Atrash, H. and Havrylyshyn, O. (1998). *Opening Up and Geographic Diversification of Trade in Transition Economies*. International Monetary Fund.

Alesina, A. and Spolaore, E. (1997). On the number and size of nations. *The Quarterly Journal of Economics*, 112(4):1027–1056.

Alesina, A. and Spolaore, E. (2005). *The Size of Nations*. MIT Press, Cambridge, MA.

Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6):2121–2168.

Baum-Snow, N., Brandt, L., Henderson, J. V., Turner, M. A., and Zhang, Q. (2017). Roads, railroads, and decentralization of Chinese cities. *Review of Economics and Statistics*, 99(3):435–448.

Becker, C., Mendelsohn, S. J., and Benderskaya, K. (2012). *Russian Urbanization in the Soviet and Post-Soviet Eras*. Human Settlements Group, International Institute for Environment and Development, London.

Behrens, K. (2024). Casualties of border changes: evidence from nighttime lights and plant exit. *Economic Policy*, 39(118):359–406.

Belkindas, M. V. and Ivanova, O. V., editors (1995). *Foreign Trade Statistics in the USSR and Successor States*. Number 18 in Studies of Economies in Transformation. The World Bank, Washington, D.C.

Blanchard, O. and Kremer, M. (1997). Disorganization. *Quarterly Journal of Economics*, 112(4):1091–1126.

Brakman, S., Garretsen, H., and Schramm, M. (2004). The strategic bombing of German cities during World War II and its impact on city growth. *Journal of Economic Geography*, 4(2):201–218.

Brinkhoff, T. (2021). City population. http://www.citypopulation.de. Accessed: 2021.

Brülhart, M., Carrère, C., and Robert-Nicoud, F. (2018). Trade and towns: Heterogeneous adjustment to a border shock. *Journal of Urban Economics*, 105:162–175.

Büchel, K. and Kyburz, S. (2020). Fast track to growth? Railway access, population growth and local displacement in 19th century Switzerland. *Journal of Economic Geography*, 20(1):155–195.

Buckley, C. (1995). The myth of managed migration: Migration control and market in the Soviet period. *Slavic Review*, 54(4):896–916.

Colella, F., Lalive, R., Sakalli, S. O., and Thoenig, M. (2019). Inference with arbitrary clustering.

Colella, F., Lalive, R., Sakalli, S. O., and Thoenig, M. (2023). acreg: Arbitrary correlation regression. *The Stata Journal*, 23(1):119–147.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45.

Council of the European Union (2009). Joint declaration of the Prague Eastern Partnership summit. 8435/09 (Presse 78).

Crozet, M. (2004). Do migrants follow market potentials? An estimation of a new economic geography model. *Journal of Economic Geography*, 4(4):439–458.

Davis, D. R. and Weinstein, D. E. (2002). Bones, bombs, and break points: The geography of economic activity. *American Economic Review*, 92(5):1269–1289.

Denisenko, M. and Chudinovskikh, O. (2017). Russia: A migration system with Soviet roots. https://www.migrationpolicy.org/article/russia-migration-system-soviet-roots. Accessed: 2023.

Dexter, K. and Rodionov, I. (2024). The factories, research and design establishments of the Soviet defence industry: A guide: Ver. 23. Technical report, University of Warwick, Department of Economics, Warwick, UK.

Dix-Carneiro, R. and Kovak, B. K. (2017). Trade liberalization and regional dynamics. *American Economic Review*, 107(10):2908–2946.

Djankov, S. and Freund, C. (2002). Trade flows in the former Soviet Union, 1987 to 1996. *Journal of Comparative Economics*, 30(1):76–90.

Donaldson, D. and Hornbeck, R. (2016). Railroads and American economic growth: A "market access" approach. *The Quarterly Journal of Economics*, 131(2):799–858.

Eaton, J. and Kortum, S. (2002). Technology, geography, and trade. *Econometrica*, 70(5):1741–1779.

Egger, P. H., Koethenbuerger, M., and Loumeau, G. (2022). Local border reforms and economic activity. *Journal of Economic Geography*, 22(1):81–102.

Eichengreen, B. (1993). A payments mechanism for the former Soviet Union: Is the EPU a relevant precedent? *Economic Policy*, 8(17):309–353.

Elvidge, C. D., Ziskin, D., Baugh, K. E., Tuttle, B. T., Ghosh, T., Pack, D. W., Erwin, E. H., and Zhizhin, M. (2009). A fifteen year record of global natural gas flaring derived from satellite data. *Energies*, 2(3):595–622.

Faber, B. (2014). Trade integration, market size, and industrialization: Evidence from China's national trunk highway system. *Review of Economic Studies*, 81(3):1046–1070.

Fujita, M., Krugman, P. R., and Venables, A. (2001). *The Spatial Economy: Cities, Regions, and International Trade*. MIT Press, Cambridge, MA.

Fukuyama, F. (1992). *The End of History and the Last Man*. Free Press, New York.

Gaidar, Y. (2010). *Collapse of an empire: Lessons for modern Russia*. Bloomsbury Publishing USA.

Gang, I. N. and Stuart, R. C. (1999). Mobility where mobility is illegal: Internal migration and city growth in the soviet union. *Journal of Population Economics*, 12(1):117–134.

Gibson, J., Olivia, S., Boe-Gibson, G., and Li, C. (2021). Which night lights data should we use in economics, and where? *Journal of Development Economics*, 149:102602.

Hanson, G. H. (2001). US–Mexico integration and regional economies: Evidence from border-city pairs. *Journal of Urban Economics*, 50(2):259–287.

Heleniak, T. (2004). Migration of the Russian diaspora after the breakup of the Soviet Union. *Journal of International Affairs*, 57(2):99–117.

Helpman, E. (1998). Topics in public economics: Theoretical and applied analysis. In *Topics in Public Economics: Theoretical and Applied Analysis*, pages 33–54. Cambridge University Press, Cambridge.

Henderson, J. V., Storeygard, A., and Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102(2):994–1028.

Hill, F. and Gaddy, C. G. (2003). *The Siberian Curse: How Communist Planners Left Russia Out in the Cold*. Brookings Institution Press, Washington, DC.

Holzman, F. D. (1985). Comecon: A "trade-destroying" customs union? *Journal of Comparative Economics*, 9(4):410–423.

IMF, World Bank, OECD, and EBRD (1991). *A Study of the Soviet Economy*, volume 1–3. IMF, World Bank, OECD, EBRD, Paris.

Jedwab, R. and Storeygard, A. (2022). The average and heterogeneous effects of transportation investments: Evidence from Sub-Saharan Africa 1960–2010. *Journal of the European Economic Association*, 20(1):1–38.

Kaminski, B. and Mitra, S. (2012). *Borderless Bazaars and Regional Integration in Central Asia: Emerging Patterns of Trade and Cross-Border Cooperation*. World Bank Publications, Washington, D.C.

Karrar, H. H. (2019). Between border and bazaar: Central Asia's informal economy. *Journal of Contemporary Asia*, 49(2):272–293.

Kofanov, D. and Mikhailova, T. (2015). Geographical concentration of Soviet industries: A comparative analysis. *Journal of the New Economic Association*, 28(4):112–141.

Kotkin, S. (2001). *Armageddon Averted: The Soviet Collapse, 1970–2000*. Oxford University Press, Oxford.

Kramer, M. and Smetana, V. (2013). *Imposing, Maintaining, and Tearing Open the Iron Curtain: The Cold War and East-Central Europe, 1945–1989*. Lexington Books, Lanham, MD.

Krugman, P. (1991). Increasing returns and economic geography. *Journal of Political Economy*, 99(3):483–499.

Li, X., Zhou, Y., Zhao, M., and Zhao, X. (2020). A harmonized global nighttime light dataset 1992–2018. *Scientific Data*, 7(1):168.

Light, M. A. (2012). What does it mean to control migration? Soviet mobility policies in comparative perspective. *Law & Social Inquiry*, 37(2):395–429.

Maurel, M. and Cheikbossian, G. (1998). The new geography of Eastern European trade. *Kyklos*, 51(1):45–72.

Meyer, B. D. (1995). Natural and quasi-experiments in economics. *Journal of Business & Economic Statistics*, 13(2):151–161.

Miguel, E. and Roland, G. (2011). The long-run impact of bombing Vietnam. *Journal of Development Economics*, 96(1):1–15.

Mukhina, I. (2007). *The Germans of the Soviet Union*. Routledge, London.

Peyrouse, S. (2008). Chinese economic presence in Kazakhstan: China's resolve and Central Asia's apprehension. *China Perspectives*, 2008(3):34–49.

Pomfret, R. (2010). Trade and transport in Central Asia. *Global Journal of Emerging Market Economies*, 2(3):237–256.

Pomfret, R. (2021). Central Asian economies: Thirty years after dissolution of the Soviet Union. *Comparative Economic Studies*, 63(4):537–556.

Radchenko, S. (2009). *Two Suns in the Heavens: The Sino-Soviet Struggle for Supremacy, 1962–1967*. Stanford University Press, Stanford, CA.

Redding, S. and Venables, A. J. (2004). Economic geography and international inequality. *Journal of International Economics*, 62(1):53–82.

Redding, S. J. and Sturm, D. M. (2008). The costs of remoteness: Evidence from German division and reunification. *American Economic Review*, 98(5):1766–1797.

Roth, J., Sant'Anna, P. H. C., Bilinski, A., and Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244.

Rubin, D. B. (1980). Randomization analysis of experimental data: The Fisher randomization test comment. *Journal of the American Statistical Association*, 75(371):591–593.

Schweiger, H., Stepanov, A., and Zacchia, P. (2022). The long-run effects of r&d place-based policies: evidence from russian science cities. *American Economic Journal: Economic Policy*, 14(3):322–351.

Smith, A. H. (1993). *Russia and the World Economy: Problems of Integration*. Routledge, London.

Spector, R. A. (2017). *Order at the Bazaar: Power and Trade in Central Asia*. Cornell University Press, Ithaca, NY.

Storeygard, A. (2016). Farther on down the road: Transport costs, trade and urban growth in Sub-Saharan Africa. *The Review of Economic Studies*, 83(3):1263–1295.

Suesse, M. (2018). Breaking the unbreakable union: Nationalism, disintegration and the Soviet economic collapse. *The Economic Journal*, 128(615):2933–2967.

# Appendix A - Figures & Tables

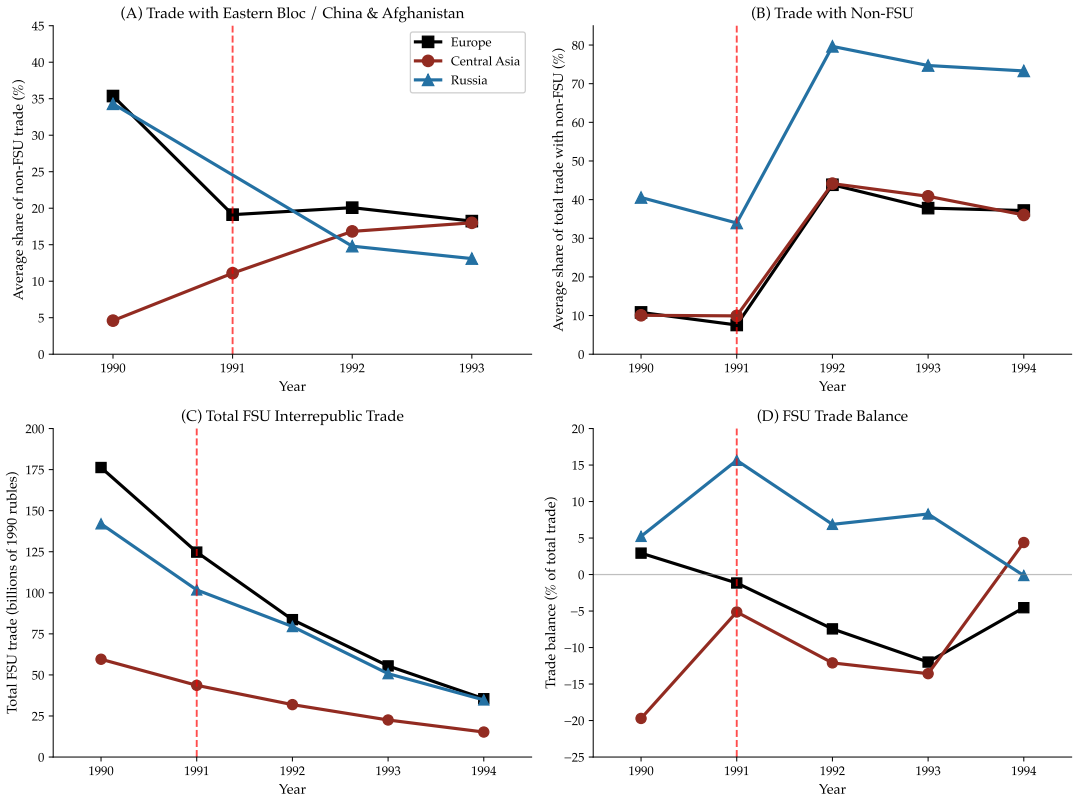



Figure A1: Post-Soviet Trade Realignment: Europe vs. Central Asia

*Notes:* This figure presents four panels illustrating trade reorientation patterns among Former Soviet Union (FSU) republics from 1990–1994. Countries are grouped into three regions: "Europe" (Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Ukraine; $n = 9$), "Central Asia" (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, Uzbekistan; $n = 5$), and Russia (shown separately). The vertical dashed line in each panel marks December 1991, when the USSR formally dissolved. The partner groups are defined by adjacency: the Eastern Bloc lay across the western frontier of the European republics, and China and Afghanistan across Central Asia's southern and eastern frontiers, so in each region the group shown is the set of non-Soviet economies whose frontier with that region changed status in 1991. Both groups cannot be shown for both regions on a consistent basis, since Belkindas and Ivanova (1995) report partner-country trade unevenly across the Central Asian republics. **Panel A** shows the average share of non-FSU trade conducted with Bulgaria, Czechoslovakia, Hungary, Poland, and Romania for European republics and Russia, and China plus Afghanistan for Central Asian republics. Turkmenistan is dropped due to lack of data. Regional values represent unweighted country averages. **Panel B** shows the average share of each region's total trade (exports plus imports) conducted with non-FSU countries (i.e., rest of world). This measure captures the overall reorientation away from intra-FSU trade toward global markets. Data are from IMF Direction of Trade Statistics and national sources. Regional values represent unweighted country averages. **Panel C** shows total interrepublic trade within the FSU, measured in billions of constant 1990 rubles to enable comparison across years despite hyperinflation that ensued. For each region, total trade equals the sum of exports and imports with all other FSU republics. **Panel D** shows the trade balance as a percentage of total FSU trade. Positive values indicate a trade surplus with other FSU republics; negative values indicate a deficit. Raw data source is Belkindas and Ivanova (1995). Coverage ends in 1994 because the Soviet-format interrepublic trade accounts on which Panels C and D rely were discontinued once the successor states began reporting as independent customs territories; no consistent constant-ruble series is available thereafter. The figure documents the sharpness and regional asymmetry of the initial reorientation rather than the trajectory of trade over the full study period.

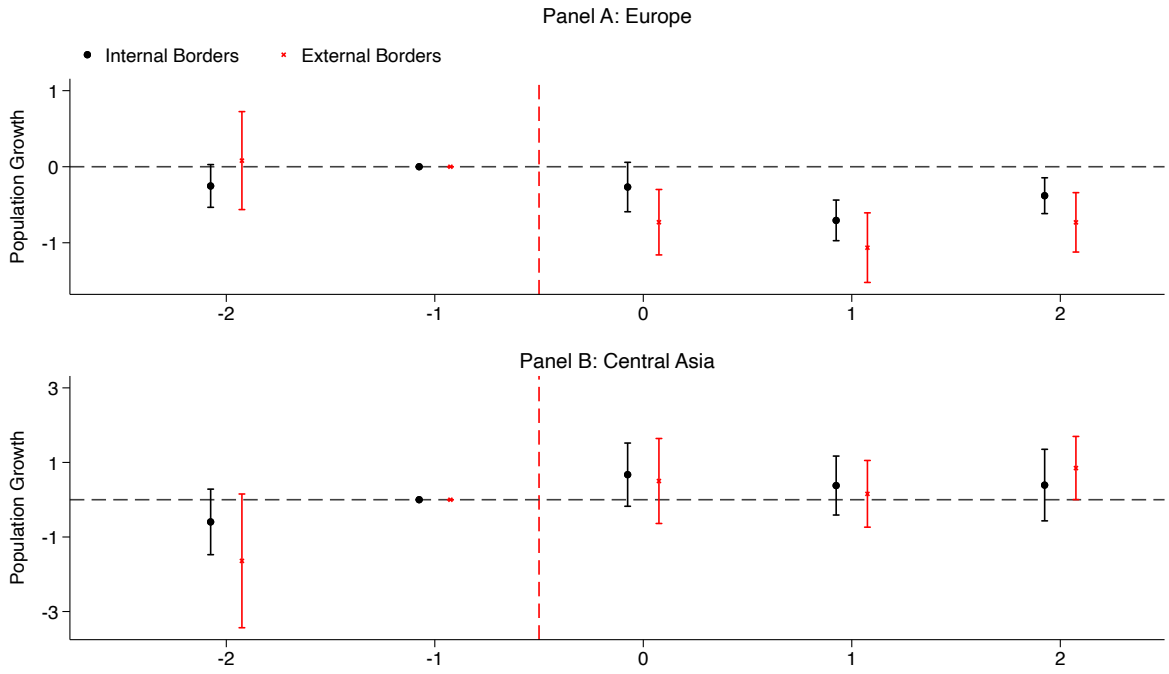
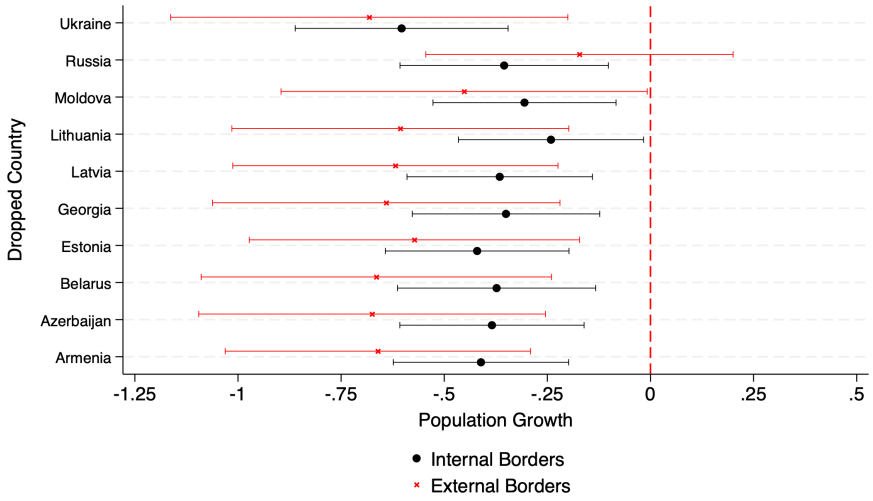


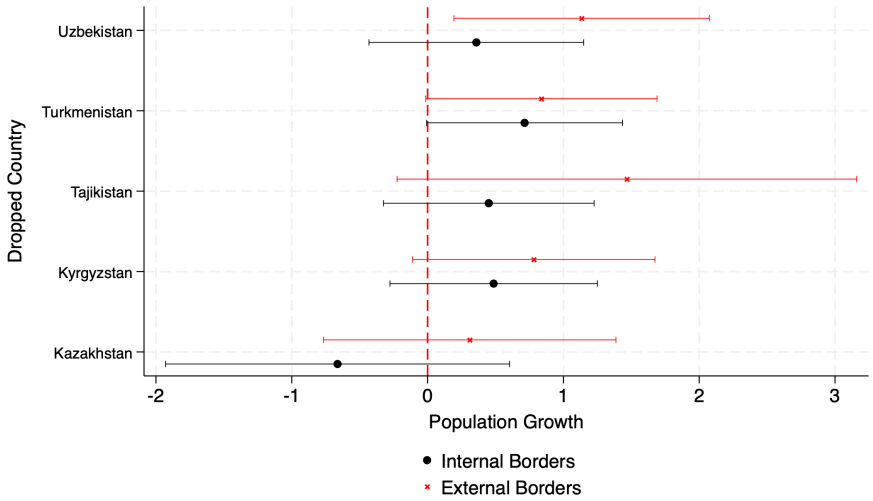

Figure A2: Event-Study Estimates: Sub-Region-by-Period Fixed Effects

*Notes: This figure replicates Figure 3 with more demanding fixed effects. Panel A replaces period fixed effects with sub-region-by-period fixed effects, where sub-regions are Baltics, Caucasus, and Eastern Europe. Panel B excludes Turkmenistan, whose census timing (1995, 1999) differs substantially from the other Central Asian countries. All other specifications are identical to Figure 3. Standard errors clustered at the city level; bars show 95% confidence intervals.*




(a) Europe

*Notes: Each point represents the estimated coefficient from a specification that drops the labeled country. Circles denote IB effects; X markers denote EB effects. Both IB and EB coefficients remain negative across all exclusions. Precision decreases when dropping Russia, which contributes the largest share of cities to the European sample. Bars show 95% confidence intervals.*




(b) Central Asia

*Notes: Each point represents the estimated coefficient from a specification that drops the labeled country. The EB coefficient remains positive in all five exclusions, though precision varies. The estimate is most attenuated when dropping Kazakhstan, which provides the largest share of interior control cities. The IB coefficient turns negative when Kazakhstan is excluded, again due to the loss in the share of interior control cities. Bars show 95% confidence intervals.*

Figure A3: Leave-One-Out Analysis

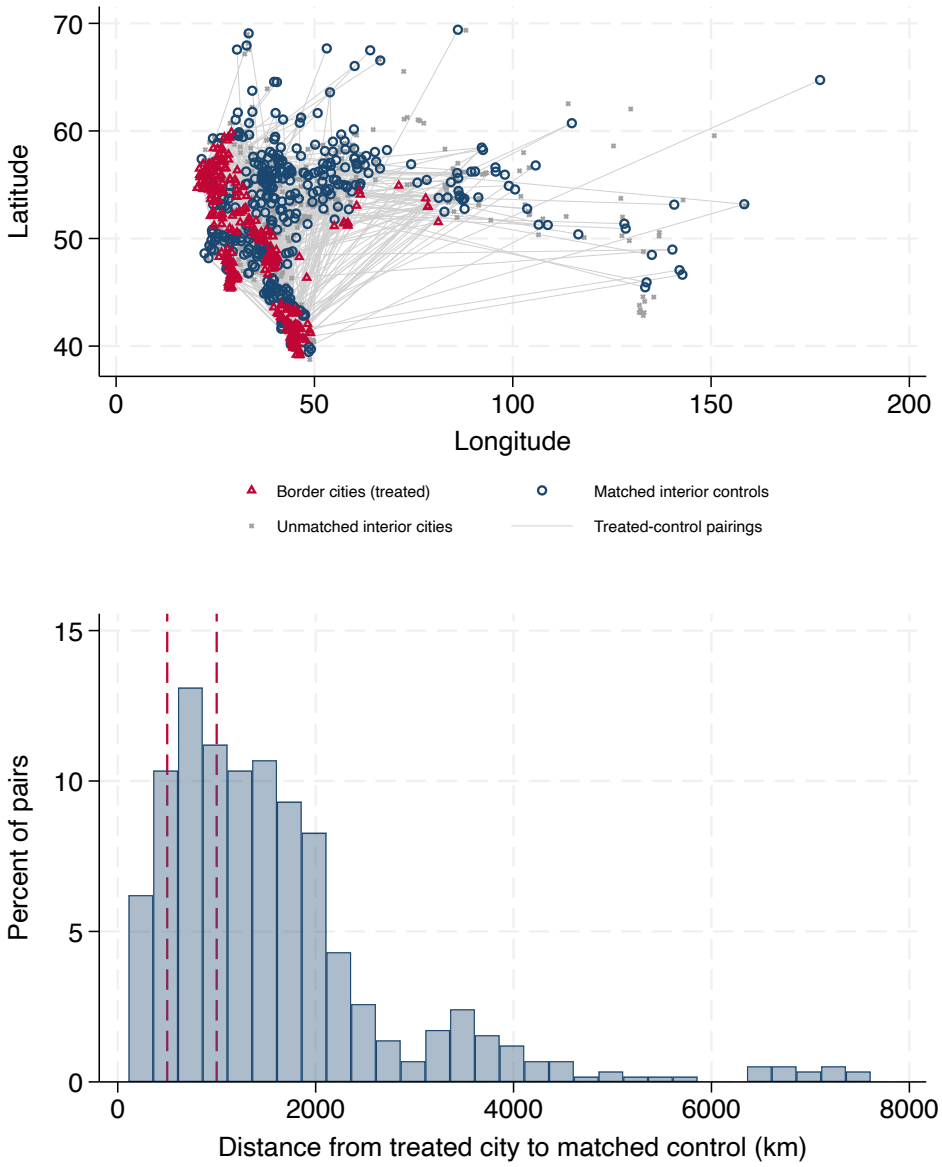



Figure A4: Matching Diagnostics: Internal Borders, Post-Soviet Europe

*Notes:* Diagnostics for the multivariate propensity score match reported in Column (1) of Table A15, which conditions on 1970 population, 1989 industrial employment, turnover, and manufacturing output, and the count and workforce of defence enterprises closing between 1989 and 1991. Matching is 1:2 nearest-neighbour with replacement under common support, with the score estimated by probit on the city cross-section. The sample is the 1,076 post-Soviet European cities with complete covariate data, fewer than the 1,221 in the baseline because the 1989 industrial data do not cover every city. **Panel A** plots each of the 290 treated border cities on support, the interior cities selected as its two matched controls, and the pairings between them; interior cities not selected as controls are shown separately. **Panel B** plots the distribution of great-circle distance between each treated city and its matched controls, with the 500 and 1,000 km radii imposed in Table A15, Columns (2)–(3) marked. The median unconstrained pair distance is 1,326 km, since the propensity score is indifferent to location; every treated city nonetheless retains two matched controls when a 500 km radius is imposed. Covariate balance is reported in Table A13.

Table A1: Summary of Census Data

| Country | Cities | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Year 6 |
|---|---|---|---|---|---|---|---|
| Armenia | 29 | 1970 | 1979 | 1989 | 2001 | 2011 | 2016 |
| Azerbaijan | 43 | 1970 | 1979 | 1989 | 2002 | 2011 | 2015 |
| Belarus | 80 | 1970 | 1979 | 1989 | 1999 | 2009 | 2019 |
| Estonia | 21 | 1970 | 1979 | 1989 | 2000 | 2011 | 2017 |
| Georgia | 41 | 1970 | 1979 | 1989 | 2002 | 2014 | 2020 |
| Kazakhstan | 77 | 1970 | 1979 | 1989 | 1999 | 2009 | 2021 |
| Kyrgyzstan | 31 | 1970 | 1979 | 1989 | 1999 | 2009 | 2021 |
| Latvia | 23 | 1970 | 1979 | 1989 | 2000 | 2011 | 2020 |
| Lithuania | 41 | 1970 | 1979 | 1989 | 2001 | 2011 | 2021 |
| Moldova | 37 | 1970 | 1979 | 1989 | 2004 | 2014 | |
| Russia | 702 | 1970 | 1979 | 1989 | 2002 | 2010 | 2021 |
| Tajikistan | 29 | 1970 | 1979 | 1989 | 2000 | 2010 | 2021 |
| Turkmenistan | 9 | 1970 | 1979 | 1989 | 1995 | 1999 | |
| Ukraine | 204 | 1970 | 1979 | 1989 | 2001 | 2014 | 2021 |
| Uzbekistan | 29 | 1970 | 1979 | 1989 | 2005 | 2020 | |
| Total | 1396 | | | | | | |

*Notes:* Census years are recorded for each country from 1970 to the most recent available year. Years 1–3 correspond to Soviet-wide censuses (1970, 1979, 1989), which are common across all countries. Years 4–6 correspond to post-independence national censuses, whose timing varies by country.

Table A2: Treatment Variable Statistics by Region

| | Mean | [SD] | Min | Max | Treated | Control | Obs. |
|---|---|---|---|---|---|---|---|
| **Panel A: Europe** | | | | | | | |
| *Outcome variable:* | | | | | | | |
| Annualized Pop. Growth | 0.583 | [1.940] | -13.61 | 27.25 | 1221 cities | | 6,004 |
| *Main treatment variables:* | | | | | | | |
| Post-1991 (Division) | 0.593 | [0.491] | 0 | 1 | 1221 | 1221 | 6,004 |
| Internal border (<75 km) | 0.281 | [0.449] | 0 | 1 | 353 | 868 | 6,004 |
| External border (<75 km) | 0.112 | [0.316] | 0 | 1 | 142 | 1079 | 6,004 |
| IB × Division | 0.163 | [0.369] | 0 | 1 | 353 | 1221 | 6,004 |
| EB × Division | 0.065 | [0.246] | 0 | 1 | 142 | 1221 | 6,004 |
| *Distance bins (mutually exclusive):* | | | | | | | |
| IB 0–25 km | 0.084 | [0.277] | 0 | 1 | 107 | 775 | 6,004 |
| IB 25–50 km | 0.111 | [0.314] | 0 | 1 | 139 | 775 | 6,004 |
| IB 50–75 km | 0.086 | [0.280] | 0 | 1 | 107 | 775 | 6,004 |
| IB 75–100 km | 0.076 | [0.264] | 0 | 1 | 93 | 775 | 6,004 |
| EB 0–25 km | 0.037 | [0.188] | 0 | 1 | 47 | 1036 | 6,004 |
| EB 25–50 km | 0.037 | [0.190] | 0 | 1 | 47 | 1036 | 6,004 |
| EB 50–75 km | 0.038 | [0.191] | 0 | 1 | 48 | 1036 | 6,004 |
| EB 75–100 km | 0.033 | [0.179] | 0 | 1 | 43 | 1036 | 6,004 |
| **Panel B: Central Asia** | | | | | | | |
| *Outcome variable:* | | | | | | | |
| Annualized Pop. Growth | 1.369 | [2.301] | -8.71 | 16.36 | 175 cities | | 805 |
| *Main treatment variables:* | | | | | | | |
| Post-1991 (Division) | 0.565 | [0.496] | 0 | 1 | 175 | 175 | 805 |
| Internal border (<75 km) | 0.591 | [0.492] | 0 | 1 | 105 | 70 | 805 |
| External border (<75 km) | 0.076 | [0.265] | 0 | 1 | 15 | 160 | 805 |
| IB × Division | 0.330 | [0.471] | 0 | 1 | 105 | 175 | 805 |
| EB × Division | 0.039 | [0.193] | 0 | 1 | 15 | 175 | 805 |
| *Distance bins (mutually exclusive):* | | | | | | | |
| IB 0–25 km | 0.323 | [0.468] | 0 | 1 | 58 | 61 | 805 |
| IB 25–50 km | 0.171 | [0.377] | 0 | 1 | 30 | 61 | 805 |
| IB 50–75 km | 0.097 | [0.296] | 0 | 1 | 17 | 61 | 805 |
| IB 75–100 km | 0.050 | [0.217] | 0 | 1 | 9 | 61 | 805 |
| EB 0–25 km | 0.045 | [0.207] | 0 | 1 | 9 | 156 | 805 |
| EB 25–50 km | 0.011 | [0.105] | 0 | 1 | 2 | 156 | 805 |
| EB 50–75 km | 0.020 | [0.140] | 0 | 1 | 4 | 156 | 805 |
| EB 75–100 km | 0.022 | [0.148] | 0 | 1 | 4 | 156 | 805 |

*Notes:* All statistics are computed on the estimation panel (city × census-period observations with non-missing annualized population growth), which matches the regression sample in Tables 1 and A6. All treatment variables are binary indicators. "Treated" and "Control" report the number of unique cities for which the indicator equals one or zero. For time-invariant indicators (e.g., Internal border), the two columns sum to the total number of cities in the region. For interaction terms (e.g., IB × Division), "Treated" counts border cities, which have the indicator equal to one only in post-1991 periods, while "Control" counts all cities that ever have the indicator equal to zero, including border cities in their pre-1991 periods; the two columns therefore overlap and do not sum to the total. For distance bins, "Control" reports interior cities (>100 km from the relevant border type), the omitted reference category in the gradient regressions (Table A6). Distance bins are mutually exclusive 25 km bands. The outcome variable row reports total unique cities in the estimation sample rather than a treated/control split.

Table A3: Descriptive Statistics by Region

| Variable | Group | Mean | SD | Min | Max | Obs. |
|---|---|---|---|---|---|---|
| | | | **Europe** | | | |
| Population | IB=1 | 76,435 | 167,631 | 3000 | 1,609,959 | 2,084 |
| | IB=0 | 116,223 | 417,391 | 3059 | 12,455,682 | 5,205 |
| 1970 Population | IB=1 | 61,305 | 136,566 | 5067 | 1,222,852 | 2,118 |
| | IB=0 | 88,995 | 299,802 | 5084 | 6,941,961 | 5,208 |
| IB Distance (km) | IB=1 | 38.48 | 20.66 | 0.080 | 74.52 | 2,118 |
| | IB=0 | 460.7 | 664.9 | 75.50 | 5078 | 5,208 |
| EB Distance (km) | IB=1 | 311.9 | 314.0 | 0.516 | 1659 | 2,118 |
| | IB=0 | 694.5 | 499.6 | 0.066 | 3034 | 5,208 |
| Large (=1) | IB=1 | 0.368 | 0.482 | 0.000 | 1.000 | 2,118 |
| | IB=0 | 0.566 | 0.496 | 0.000 | 1.000 | 5,208 |
| Population | EB=1 | 66,605 | 140,177 | 3000 | 1,201,539 | 826 |
| | EB=0 | 109,734 | 383,409 | 3059 | 12,455,682 | 6,463 |
| 1970 Population | EB=1 | 49,120 | 103,983 | 5067 | 766,705 | 852 |
| | EB=0 | 85,184 | 277,502 | 5084 | 6,941,961 | 6,474 |
| IB Distance (km) | EB=1 | 341.9 | 823.6 | 0.080 | 3334 | 852 |
| | EB=0 | 338.2 | 555.1 | 0.259 | 5078 | 6,474 |
| EB Distance (km) | EB=1 | 37.47 | 22.33 | 0.066 | 74.85 | 852 |
| | EB=0 | 655.8 | 471.7 | 75.36 | 3034 | 6,474 |
| Large (=1) | EB=1 | 0.282 | 0.450 | 0.000 | 1.000 | 852 |
| | EB=0 | 0.538 | 0.499 | 0.000 | 1.000 | 6,474 |
| | | | **Central Asia** | | | |
| Population | IB=1 | 108,327 | 252,647 | 4351 | 2,571,668 | 581 |
| | IB=0 | 81,422 | 114,737 | 6010 | 1,184,469 | 399 |
| 1970 Population | IB=1 | 70,553 | 162,565 | 5053 | 1,384,509 | 630 |
| | IB=0 | 55,322 | 76,512 | 6010 | 523,271 | 420 |
| IB Distance (km) | IB=1 | 25.86 | 19.86 | 0.655 | 70.68 | 630 |
| | IB=0 | 204.5 | 108.2 | 78.52 | 502 | 420 |
| EB Distance (km) | IB=1 | 298.0 | 280.8 | 6.290 | 1540 | 630 |
| | IB=0 | 560.8 | 409.1 | 2.824 | 1464 | 420 |
| Large (=1) | IB=1 | 0.390 | 0.488 | 0.000 | 1.000 | 630 |
| | IB=0 | 0.514 | 0.500 | 0.000 | 1.000 | 420 |
| Population | EB=1 | 55,059 | 107,879 | 5053 | 604,700 | 76 |
| | EB=0 | 100,930 | 214,151 | 4351 | 2,571,668 | 904 |
| 1970 Population | EB=1 | 32,674 | 60,280 | 5053 | 253,118 | 90 |
| | EB=0 | 67,441 | 139,683 | 5239 | 1,384,509 | 960 |
| IB Distance (km) | EB=1 | 134.1 | 100.0 | 26.04 | 382 | 90 |
| | EB=0 | 93.88 | 112.7 | 0.655 | 502 | 960 |
| EB Distance (km) | EB=1 | 26.35 | 21.04 | 2.824 | 59.52 | 90 |
| | EB=0 | 438.5 | 358.3 | 80.53 | 1540 | 960 |
| Large (=1) | EB=1 | 0.267 | 0.445 | 0.000 | 1.000 | 90 |
| | EB=0 | 0.456 | 0.498 | 0.000 | 1.000 | 960 |
| | | | **Europe** | | | |
| Log(Lights) | IB=1 | 7.524 | 1.735 | 0.693 | 11.95 | 6,072 |
| | IB=0 | 8.640 | 1.480 | 0.693 | 14.50 | 13,332 |
| Log(Lights) | EB=1 | 7.177 | 1.588 | 0.693 | 11.13 | 2,662 |
| | EB=0 | 8.468 | 1.586 | 0.693 | 14.50 | 16,742 |
| | | | **Central Asia** | | | |
| Log(Lights) | IB=1 | 8.105 | 1.751 | 0.693 | 12.05 | 1,496 |
| | IB=0 | 8.134 | 1.616 | 3.178 | 11.54 | 1,166 |
| Log(Lights) | EB=1 | 6.945 | 1.720 | 3.178 | 11.50 | 308 |
| | EB=0 | 8.271 | 1.628 | 0.693 | 12.05 | 2,354 |

*Notes:* Statistics are reported on the full city × census-period panel, which includes six periods per city (three pre-1991, three post-1991). Population is the level recorded at each census. 1970 Population is time-invariant but reported for all city-periods. IB and EB Distance measure straight-line kilometers to the nearest internal and external post-Soviet border. Large is a binary indicator for cities above the 1970 sample median population. Log(Lights) is from a separate annual city-year panel (1992–2013) matched to the same city universe; observation counts reflect city × year. Treatment variable statistics and the outcome variable (annualized population growth) are reported in Table A3, computed on the estimation sample (city-period observations with non-missing growth rates).

## Table A4: Border Effects by Type of Exposure

| | Dependent Variable: Population Growth | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| IB only × Division | −0.358*** | −0.172 | 0.552 | 0.778** |
| | (0.113) | (0.121) | (0.395) | (0.393) |
| EB only × Division | −0.441 | −0.437 | 1.356** | 1.192* |
| | (0.291) | (0.305) | (0.620) | (0.629) |
| Both IB & EB × Division | −1.099*** | −1.047*** | 0.803 | 1.078** |
| | (0.266) | (0.244) | (0.520) | (0.518) |
| Both − IB only | −0.740 | −0.875 | 0.251 | 0.301 |
| *p*-value | 0.007 | 0.001 | 0.564 | 0.492 |
| Both − EB only | −0.657 | −0.610 | −0.553 | −0.113 |
| *p*-value | 0.088 | 0.105 | 0.388 | 0.862 |
| *p*: effects additive | 0.456 | 0.279 | 0.143 | 0.242 |
| Outcome mean (%) | 0.583 | 0.583 | 1.369 | 1.309 |
| Sample | Europe | Europe | CA | CA (exc. Turk.) |
| Cities | 1,221 | 1,221 | 175 | 166 |
| Observations | 6,004 | 6,004 | 805 | 769 |
| City FE | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | | ✓ | ✓ |
| Sub-Region-by-Period FE | | ✓ | | |

*Notes:* The dependent variable is annualized city population growth (%). Cities are partitioned into four mutually exclusive categories by exposure: within 75 km of an internal border only ("IB only"), within 75 km of an external border only ("EB only"), within 75 km of both ("Both"), and interior cities beyond 75 km of any border, which form the omitted reference group. "Division" equals one for post-1991 periods. Because the three indicators are mutually exclusive, each coefficient reports the effect of that exposure category relative to interior cities. "Both − IB only" and "Both − EB only" report the incremental effect of adding the second type of border exposure, with *p*-values from Wald tests of the linear combination. The additivity row reports the *p*-value from a test that the dual effect equals the sum of the two single-exposure effects; equivalently, it is the *p*-value on a dual-exposure interaction added to a specification containing both border indicators. Of the 80 dual-exposure cities in post-Soviet Europe, 22 are in Armenia and 23 in Moldova. Columns (1)–(2) estimate the decomposition for post-Soviet Europe, with Column (2) replacing period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, & Eastern Europe). Columns (3)–(4) estimate it for Central Asia, with Column (4) excluding Turkmenistan, whose census timing differs from the other Central Asian countries; sub-region-by-period fixed effects are not reported for Central Asia because it constitutes a single sub-region. In post-Soviet Europe, 273 cities are near an internal border only, 62 near an external border only, 80 near both, and 806 are interior. The corresponding counts for Central Asia are 100, 10, 5, and 60. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A5: Border Effects by City Size

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | All | | Europe | | Central Asia | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| IB × Division | −0.197 | −0.024 | −0.447** | −0.130 | 0.923 | 0.180 |
| | (0.170) | (0.105) | (0.177) | (0.115) | (0.611) | (0.405) |
| EB × Division | −0.567** | −0.187 | −0.587** | −0.200 | 1.402** | 0.601 |
| | (0.257) | (0.203) | (0.273) | (0.205) | (0.629) | (0.803) |
| City size | Small | Large | Small | Large | Small | Large |
| Outcome mean (%) | 0.838 | 0.517 | 0.752 | 0.423 | 1.394 | 1.337 |
| Treated IB cities | 287 | 171 | 223 | 130 | 64 | 41 |
| Treated EB cities | 113 | 44 | 102 | 40 | 11 | 4 |
| Cities | 698 | 698 | 600 | 621 | 98 | 77 |
| Observations | 3,372 | 3,437 | 2,919 | 3,085 | 453 | 352 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders, respectively; "Division" equals one for post-1991 periods. Cities are split at the median 1970 population across the full sample (approximately 28,800 residents), so the cut point is common to both regions rather than computed within region. This absolute threshold is the relevant one for assessing whether the effects are concentrated in the large cities that absorbed the bulk of post-1991 return migration. The pooled cut divides each region close to evenly, placing 49 percent of European and 56 percent of Central Asian cities in the small group.. Because the split partitions treated and control cities together, each column compares border to interior cities within the same size range. Odd-numbered columns report small cities and even-numbered columns large cities, for the pooled sample (Columns 1–2), post-Soviet Europe (Columns 3–4), and Central Asia (Columns 5–6). "Treated IB cities" and "Treated EB cities" report the number of unique cities within 75 km of the respective border type in each estimation sample; these need not sum to the total number of border cities reported elsewhere because a city may lie near both border types. Column (6) contains four treated external border cities and should be read as descriptive. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A6: Coastal Access and Border Effects (Post-Soviet Europe)

| | Dependent Variable: Population Growth | | | | | | |
|---|---|---|---|---|---|---|---|
| | Open sea, 75 km | | | | Open sea, 25 km | Incl. Caspian | Baltic & Black Sea |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| IB × Division | −0.397*** | −0.240** | −0.368*** | | −0.398*** | −0.402*** | −0.406*** |
| | (0.111) | (0.114) | (0.123) | | (0.111) | (0.111) | (0.111) |
| EB × Division | −0.592*** | −0.664*** | −0.784*** | | −0.592*** | −0.600*** | −0.595*** |
| | (0.201) | (0.196) | (0.227) | | (0.202) | (0.201) | (0.200) |
| IB × Division × Coastal | | | | −0.383 | | | |
| | | | | (0.240) | | | |
| IB × Division × Inland | | | | −0.366*** | | | |
| | | | | (0.123) | | | |
| EB × Division × Coastal | | | | 0.377 | | | |
| | | | | (0.366) | | | |
| EB × Division × Inland | | | | −0.782*** | | | |
| | | | | (0.227) | | | |
| Coastal × Division | −0.132 | 0.044 | | −0.282* | −0.445** | 0.101 | 0.068 |
| | (0.134) | (0.142) | | (0.167) | (0.173) | (0.130) | (0.134) |
| $p$: IB coastal = inland | | | | 0.951 | | | |
| $p$: EB coastal = inland | | | | 0.007 | | | |
| Coastal cities | 164 | 164 | | 164 | 91 | 201 | 136 |
| of which near EB | 23 | 23 | | 23 | 12 | 30 | 19 |
| Outcome mean (%) | 0.583 | 0.583 | 0.610 | 0.583 | 0.583 | 0.583 | 0.583 |
| City sample | All | All | Inland | All | All | All | All |
| Cities | 1,221 | 1,221 | 1,057 | 1,221 | 1,221 | 1,221 | 1,221 |
| Observations | 6,004 | 6,004 | 5,217 | 6,004 | 6,004 | 6,004 | 6,004 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Sub-Region-by-Period FE | | ✓ | | | | | |

*Notes:* The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders, respectively; "Division" equals one for post-1991 periods. "Coastal" equals one for cities within the stated distance of the coastline, measured as straight-line distance. Columns (1)–(4) use the open-sea coastline within 75 km, which excludes the Caspian, a landlocked sea with no direct sea route to world markets; the Caspian accounts for 37 European cities, mostly in Azerbaijan, Dagestan, and Astrakhan. Column (1) adds a coastal trajectory to the baseline specification, and Column (2) replaces period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Slavic Europe plus Moldova). Column (3) drops all coastal cities, so that neither treated nor control cities have direct access to the sea. Column (4) allows the border effects to differ by coastal status; because the coastal main effect is included, each border coefficient compares border cities with interior cities of the same coastal status, and the $p$-values are from Wald tests that the coastal and inland coefficients are equal. Column (5) narrows the coastal definition to 25 km, closer to direct port access. Column (6) restores the Caspian. Column (7) restricts the coast to the Baltic, Black, and Azov Seas, excluding the Arctic and Pacific coasts, where coastal status coincides with the withdrawal of northern and Far Eastern subsidies and with naval bases. "Coastal cities" and "of which near EB" report unique cities in each estimation sample. Central Asia is omitted because its only coastline is the Caspian. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A7: Robustness Checks

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A: Europe** | | | | | | |
| IB × Division | −0.399*** | −0.396*** | −0.359*** | −0.690*** | −0.532*** | −0.529*** |
| | (0.114) | (0.110) | (0.113) | (0.121) | (0.108) | (0.118) |
| EB × Division | −0.587*** | −0.648*** | −0.442 | −0.412* | −0.534*** | −0.392** |
| | (0.204) | (0.194) | (0.291) | (0.228) | (0.180) | (0.179) |
| City sample | Pop 1970 <500k | Excl. 1989–99 | Excl. both IB&EB | Non-conflict | Excl. Asian RU | Dist <750 km |
| Outcome mean (%) | 0.582 | 0.786 | 0.584 | 0.626 | 0.601 | 0.639 |
| Cities | 1,191 | 1,221 | 1,141 | 1,049 | 1,028 | 771 |
| Observations | 5,856 | 4,783 | 5,629 | 5,193 | 5,039 | 3,754 |
| **Panel B: Central Asia** | | | | | | |
| IB × Division | 0.464 | 0.466 | 0.552 | 0.495 | 0.569 | 0.456 |
| | (0.376) | (0.376) | (0.395) | (0.397) | (0.378) | (0.450) |
| EB × Division | 0.970** | 0.755* | 1.356** | 1.442* | 0.938** | 0.877* |
| | (0.453) | (0.398) | (0.620) | (0.866) | (0.457) | (0.485) |
| City sample | Pop 1970 <500k | Excl. 1989–99 | Excl. both IB&EB | Non-conflict | Excl. KZ–RU IB | Dist <750 km |
| Outcome mean (%) | 1.374 | 1.777 | 1.349 | 1.362 | 1.413 | 1.412 |
| Cities | 172 | 175 | 170 | 155 | 165 | 144 |
| Observations | 791 | 630 | 785 | 722 | 755 | 650 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* The dependent variable is annualized city population growth (%). Column (1) excludes cities above 500,000 population in 1970, where *propiska* restrictions were most stringent. Column (2) excludes the 1989–1999 intercensal period. Column (3) excludes cities within 75 km of both internal and external borders; because these cities are dropped rather than assigned a separate indicator, its coefficients reproduce the internal-only and external-only estimates in Table A4. Column (4) drops cities near conflict-affected border pairs (Armenia–Azerbaijan, Georgia–Russia, Moldova–Ukraine, Russia–Ukraine, and Kyrgyzstan–Tajikistan for internal; Armenia–Turkey, Afghanistan–Tajikistan for external). Column (5) applies a region-specific restriction. In Panel A it excludes the 193 Russian cities in the Siberian, Far Eastern, and Urals federal districts, confining the Russian sample to districts west of the Urals and testing whether the classification of Russia with post-Soviet Europe drives the European estimates. In Panel B it excludes the ten Central Asian cities within 75 km of the Kazakhstan–Russia frontier, the one Central Asian internal border that severed a dense industrial corridor. Column (6) restricts the control group to cities within 750 km of any border, to test whether the effects are driven by contrasts with very remote interior cities. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Table A8: Market Access and Urban Growth: Robustness to Spatial Autocorrelation**

| | Dependent Variable: Population Growth | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Panel A: Europe** | | | | | |
| IB × Division | −0.402*** | −0.402*** | −0.402*** | −0.402*** | −0.402** |
| | (0.106) | (0.122) | (0.136) | (0.146) | (0.169) |
| EB × Division | −0.595*** | −0.595*** | −0.595*** | −0.595*** | −0.595*** |
| | (0.184) | (0.203) | (0.209) | (0.203) | (0.198) |
| Observations | 6,004 | 6,004 | 6,004 | 6,004 | 6,004 |
| **Panel B: Central Asia** | | | | | |
| IB × Division | 0.471* | 0.471* | 0.471* | 0.471* | 0.471* |
| | (0.282) | (0.279) | (0.280) | (0.283) | (0.275) |
| EB × Division | 0.969** | 0.969** | 0.969** | 0.969** | 0.969** |
| | (0.423) | (0.424) | (0.434) | (0.483) | (0.470) |
| Distance cutoff | 25 km | 50 km | 75 km | 100 km | 200 km |
| Observations | 805 | 805 | 805 | 805 | 805 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* The dependent variable is annualized city population growth (%). Each column reports the baseline specification (city and period fixed effects) with Conley spatial standard errors (Conley, 1999) computed using the `acreg` package (Colella et al., 2019). Columns vary the distance cutoff beyond which spatial correlation is assumed to be zero. Point estimates are identical across columns; only standard errors change. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A9: Distance Gradient: Border Proximity and Urban Growth

| | Dependent Variable: Population Growth | | | |
|---|---|---|---|---|
| | Europe | | Central Asia | |
| | (1) | (2) | (3) | (4) |
| IB (0–25 km) × Division | −0.486*** | −0.264 | 0.633 | 0.886** |
| | (0.170) | (0.171) | (0.434) | (0.436) |
| IB (25–50 km) × Division | −0.363** | −0.195 | 0.488 | 0.768 |
| | (0.155) | (0.149) | (0.614) | (0.631) |
| IB (50–75 km) × Division | −0.225 | −0.118 | 0.045 | 0.374 |
| | (0.196) | (0.197) | (0.574) | (0.579) |
| IB (75–100 km) × Division | 0.105 | 0.184 | −0.233 | 0.200 |
| | (0.168) | (0.167) | (0.475) | (0.485) |
| EB (0–25 km) × Division | −1.055*** | −1.224*** | 1.438** | 1.362** |
| | (0.360) | (0.372) | (0.626) | (0.630) |
| EB (25–50 km) × Division | −0.377 | −0.447 | 0.148 | 0.440 |
| | (0.384) | (0.363) | (0.391) | (0.392) |
| EB (50–75 km) × Division | −0.485* | −0.561** | 0.755 | 0.084 |
| | (0.258) | (0.254) | (0.636) | (0.337) |
| EB (75–100 km) × Division | −0.835*** | −0.925*** | 0.953 | 0.257 |
| | (0.206) | (0.224) | (1.126) | (0.982) |
| Outcome mean (%) | 0.583 | 0.583 | 1.369 | 1.309 |
| Cities | 1,221 | 1,221 | 175 | 166 |
| Observations | 6,004 | 6,004 | 805 | 769 |
| City FE | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | | ✓ | ✓ |
| Sub-Region-by-Period FE | | ✓ | | |

*Notes:* The dependent variable is annualized city population growth (%). Each row reports the coefficient on the interaction between a distance-bin indicator and a post-1991 indicator. The omitted reference category is cities beyond 100 km from any border. Column (1) is the baseline for Europe with city and period fixed effects. Column (2) replaces period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Slavic Europe plus Moldova). Column (3) is the baseline for Central Asia. Column (4) excludes Turkmenistan. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A10: Alternative Distance Thresholds for Border Proximity

| | Dependent Variable: Population Growth | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Distance threshold | 25 km | 50 km | 75 km | 100 km | 125 km |
| **Panel A: Europe** | | | | | |
| IB × Division | −0.537*** | −0.442*** | −0.402*** | −0.255** | −0.178* |
| | (0.167) | (0.121) | (0.111) | (0.105) | (0.102) |
| EB × Division | −1.051*** | −0.676** | −0.595*** | −0.713*** | −0.833*** |
| | (0.353) | (0.267) | (0.200) | (0.168) | (0.149) |
| Treated IB cities | 107 | 246 | 353 | 446 | 528 |
| Treated EB cities | 47 | 94 | 142 | 185 | 225 |
| Interior cities | 1,074 | 921 | 806 | 709 | 626 |
| Outcome mean (%) | | | 0.583 | | |
| Cities | | | 1,221 | | |
| Observations | | | 6,004 | | |
| **Panel B: Central Asia** | | | | | |
| IB × Division | 0.457 | 0.544 | 0.471 | 0.427 | 0.457 |
| | (0.319) | (0.347) | (0.370) | (0.403) | (0.442) |
| EB × Division | 1.374** | 1.123** | 0.969** | 0.928** | 0.542 |
| | (0.590) | (0.525) | (0.451) | (0.457) | (0.397) |
| Treated IB cities | 58 | 88 | 105 | 114 | 124 |
| Treated EB cities | 9 | 11 | 15 | 19 | 25 |
| Interior cities | 108 | 78 | 60 | 49 | 40 |
| Outcome mean (%) | | | 1.369 | | |
| Cities | | | 175 | | |
| Observations | | | 805 | | |
| City FE | | | ✓ | | |
| Period FE | | | ✓ | | |

*Notes:* The dependent variable is annualized city population growth (%). Each column redefines both treatments at the stated distance threshold: a city is treated if it lies within that distance of the relevant frontier, and the control group is cities beyond it. Column (3) reproduces the baseline 75 km specification. The estimation sample is held fixed across columns, so only the allocation of cities between treated and control groups changes; "Treated IB" and "Treated EB" overlap for cities near both frontiers, so the three counts do not sum to the sample total. Internal border effects attenuate monotonically as the threshold widens, which is what a frontier-specific shock implies, since wider rings add cities with weaker exposure. External border effects do not, consistent with the broader peripherality component discussed in the text and in Table A11. In Panel B the interior control group falls from 108 cities at 25 km to 40 at 125 km, so the widest Central Asian columns rest on a thin comparison group. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A11: Summary Statistics: Sectoral Composition (1989)

| | IB Regression Sample | | | | | EB Regression Sample | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | Med. | SD | Min | Max | Mean | Med. | SD | Min | Max |
| **Panel A: Europe** | | | | | | | | | | |
| Manufacturing share | 0.834 | 0.925 | 0.234 | 0 | 1 | 0.841 | 0.924 | 0.224 | 0 | 1 |
| Agriculture share | 0.026 | 0 | 0.088 | 0 | 1 | 0.022 | 0 | 0.068 | 0 | 0.813 |
| Mining share | 0.074 | 0 | 0.204 | 0 | 1 | 0.069 | 0 | 0.196 | 0 | 1 |
| Total empl. (000s) | 17.3 | 5.381 | 40.6 | 0.022 | 785 | 17.2 | 5.653 | 41.5 | 0.022 | 785 |
| (total) | 1021 | | | | | 848 | | | | |
| of which: IB cities | 290 (0.284) | | | | | 62 (0.073) | | | | |
| of which: EB cities | 62 (0.061) | | | | | 117 (0.138) | | | | |
| of which: Interior | 731 (0.716) | | | | | 731 (0.862) | | | | |
| *Heterogeneity diagnostics (treated cities):* | | | | | | | | | | |
| Manuf. quartile (Q1/Q2/Q3/Q4) | 249(77)/258(59)/255(64)/259(90) | | | | | 203(31)/221(22)/220(29)/204(35) | | | | |
| Manuf. median split (High/Low) | 514(154) / 507(136) | | | | | 424(64) / 424(53) | | | | |
| Agric. median split (High/Low) | 494(127) / 527(163) | | | | | 414(47) / 434(70) | | | | |
| Mining median split (High/Low) | 294(90) / 727(200) | | | | | 240(36) / 608(81) | | | | |
| **Panel B: Central Asia** | | | | | | | | | | |
| Manufacturing share | 0.753 | 0.849 | 0.281 | 0 | 1 | 0.686 | 0.755 | 0.286 | 0 | 1 |
| Agriculture share | 0.036 | 0 | 0.111 | 0 | 1 | 0.074 | 0.001 | 0.17 | 0 | 1 |
| Mining share | 0.111 | 0 | 0.253 | 0 | 1 | 0.102 | 0 | 0.225 | 0 | 0.93 |
| Total empl. (000s) | 12.1 | 4.728 | 20.5 | 0.026 | 143 | 10.1 | 3.03 | 18.3 | 0.118 | 118 |
| (total) | 141 | | | | | 61 | | | | |
| of which: IB cities | 91 (0.645) | | | | | 3 (0.049) | | | | |
| of which: EB cities | 3 (0.021) | | | | | 11 (0.18) | | | | |
| of which: Interior | 50 (0.355) | | | | | 50 (0.82) | | | | |
| *Heterogeneity diagnostics (treated cities):* | | | | | | | | | | |
| Manuf. median split (High/Low) | 72(55) / 69(36) | | | | | 19(2) / 42(9) | | | | |
| Agric. median split (High/Low) | 69(45) / 72(46) | | | | | 32(8) / 29(3) | | | | |
| Mining median split (High/Low) | 45(27) / 96(64) | | | | | 19(1) / 42(10) | | | | |

*Notes:* Sectoral shares are calculated as sector employment divided by total city employment in 1989. The IB regression sample includes all cities except those near only external borders, matching internal border specifications (Tables 1–2). The EB regression sample includes all cities except those near only internal borders. "Of which" rows report the count and share of each city type; IB and EB counts need not sum to the total because some cities fall within 75 km of both border types. In the heterogeneity diagnostics, numbers in parentheses indicate treated (border) cities; the remainder are interior controls. "Manuf. quartile" reports the Q1/Q2/Q3/Q4 distribution across manufacturing quartiles used in Table 2, Columns (1)–(2). "Median split (High/Low)" reports cell sizes for specifications in Tables 2 and A8. In both regions, the median agriculture and mining shares are zero, so "High" captures cities with any positive employment in that sector.

**Table A12: Urban Growth and Sectoral Composition: Additional Median Splits**

| | Dependent Variable: Population Growth | | | |
|---|---|---|---|---|
| | Internal Border | | External Border | |
| | (1) Agric. | (2) Mining | (3) Agric. | (4) Mining |
| **Panel A: Europe** | | | | |
| Border × Div. × High Agric. | −0.307* | | −0.584 | |
| | (0.165) | | (0.387) | |
| Border × Div. × Low Agric. | −0.446** | | −0.758** | |
| | (0.181) | | (0.295) | |
| Border × Div. × High Mining | | 0.059 | | −0.170 |
| | | (0.196) | | (0.314) |
| Border × Div. × Low Mining | | −0.578*** | | −0.915*** |
| | | (0.155) | | (0.305) |
| High Agric. × Div. | 0.005 | | 0.005 | |
| | (0.126) | | (0.126) | |
| High Mining × Div. | | −0.215 | | −0.215 |
| | | (0.146) | | (0.146) |
| $p$-value: High = Low | 0.570 | 0.011 | 0.720 | 0.090 |
| Outcome mean (%) | 0.527 | 0.527 | 0.578 | 0.578 |
| Cities | 1,021 | 1,021 | 848 | 848 |
| Observations | 5,033 | 5,033 | 4,206 | 4,206 |
| **Panel B: Central Asia** | | | | |
| | (1) Manuf. | (2) Mining | (3) Manuf. | (4) Mining |
| Border × Div. × High Manuf. | 0.197 | | 1.483 | |
| | (0.529) | | (1.291) | |
| Border × Div. × Low Manuf. | 0.499 | | 1.479* | |
| | (0.604) | | (0.778) | |
| Border × Div. × High Mining | | 1.470* | | 1.702** |
| | | (0.859) | | (0.840) |
| Border × Div. × Low Mining | | −0.133 | | 0.915 |
| | | (0.445) | | (0.652) |
| High Manuf. × Div. | 0.476 | | 0.475 | |
| | (0.711) | | (0.717) | |
| High Mining × Div. | | −1.577* | | −1.577* |
| | | (0.894) | | (0.902) |
| $p$-value: High = Low | 0.705 | 0.099 | 0.997 | 0.453 |
| Outcome mean (%) | 1.362 | 1.362 | 1.328 | 1.328 |
| Cities | 141 | 141 | 61 | 61 |
| Observations | 652 | 652 | 284 | 284 |
| City FE | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ |

*Notes:* The dependent variable is annualized city population growth (%). Cities are split at the within-region median of the relevant 1989 employment share; "High" denotes cities above the median and "Low" denotes cities at or below the median. For agriculture and mining in Europe, where the median share is zero, the split compares cities with any positive employment in that sector to those without. Panel A reports Europe, excluding manufacturing (shown in the main table as quartiles). Panel B reports Central Asia, excluding agriculture (shown in the main table as a median split). "Border" refers to cities within 75 km of an internal border in Columns (1)–(2) and an external border in Columns (3)–(4). "Div." equals one for post-1991 periods. The Central Asian external border sample contains 61 clusters, of which 11 are near external borders; sectoral splits yield small cells, so Columns (3)–(4) of Panel B should be interpreted with caution. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A13: Covariate Balance in Matched Samples

| | Treated (Border) | | | | | Control (Interior) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Matching variable | Mean | SD | Min | Max | *N* | Mean | SD | Min | Max | *N* |
| **Panel A: Europe** | | | | | | | | | | |
| *Internal border matching:* | | | | | | | | | | |
| 1970 Population | 61,465 | 136,889 | 5,235 | 1,222,852 | 352 | 70,598 | 140,246 | 5,226 | 1,170,133 | 415 |
| 1989 Ind. Employment | 15,564 | 31,924 | 54.0 | 250,404 | 289 | 15,634 | 30,281 | 24.0 | 274,489 | 399 |
| 1989 Ind. Turnover | 340,085 | 644,501 | 70.0 | 4,758,280 | 290 | 370,136 | 696,305 | 64.0 | 4,796,282 | 379 |
| 1989 Manufacturing | 216,068 | 463,404 | 0.000 | 3,781,983 | 290 | 259,122 | 514,307 | 0.000 | 3,728,328 | 383 |
| 1989 SIC | 13,427 | 6,547 | 2,730 | 32,720 | 279 | 12,564 | 6,432 | 2,730 | 32,720 | 719 |
| Military count | 8.084 | 17.8 | 1.000 | 160.0 | 179 | 18.6 | 29.5 | 1.000 | 208.0 | 94 |
| Military size | 17.6 | 38.0 | 2.000 | 339.0 | 179 | 40.6 | 63.0 | 2.000 | 443.0 | 94 |
| *External border matching:* | | | | | | | | | | |
| 1970 Population | 49,433 | 104,595 | 5,235 | 766,705 | 141 | 53,785 | 108,825 | 5,226 | 770,905 | 238 |
| 1989 Ind. Employment | 12,137 | 24,526 | 54.0 | 161,117 | 115 | 13,034 | 25,561 | 59.0 | 158,143 | 199 |
| 1989 Ind. Turnover | 256,859 | 504,018 | 70.0 | 3,176,533 | 117 | 254,788 | 505,249 | 64.0 | 3,107,729 | 207 |
| 1989 Manufacturing | 159,088 | 326,866 | 0.000 | 2,117,026 | 117 | 168,095 | 339,349 | 0.000 | 2,161,119 | 198 |
| 1989 SIC | 13,673 | 6,316 | 2,730 | 27,520 | 115 | 12,196 | 6,151 | 2,730 | 27,520 | 826 |
| Military count | 7.758 | 15.4 | 1.000 | 105.0 | 66 | 14.7 | 28.1 | 1.000 | 166.0 | 49 |
| Military size | 16.5 | 32.5 | 2.000 | 218.0 | 66 | 31.7 | 60.1 | 2.000 | 360.0 | 49 |
| **Panel B: Central Asia** | | | | | | | | | | |
| *Internal border matching:* | | | | | | | | | | |
| 1970 Population | 54,241 | 76,263 | 6,197 | 430,618 | 97 | 57,006 | 81,603 | 6,010 | 523,271 | 62 |
| 1989 Ind. Employment | 11,542 | 16,064 | 129.0 | 87,890 | 86 | 11,075 | 19,623 | 118.0 | 117,556 | 52 |
| 1989 Ind. Turnover | 312,309 | 448,680 | 1,330 | 2,403,345 | 88 | 456,404 | 672,555 | 954.0 | 2,731,450 | 48 |
| 1989 Manufacturing | 191,509 | 340,011 | 0.000 | 1,825,853 | 91 | 187,719 | 319,809 | 0.000 | 1,414,383 | 49 |
| 1989 SIC | 10,820 | 6,123 | 2,730 | 23,000 | 69 | 11,694 | 6,408 | 2,730 | 23,000 | 42 |
| Military count | 3.209 | 4.877 | 1.000 | 28.0 | 43 | 3.200 | 2.484 | 1.000 | 10.0 | 15 |
| Military size | 7.047 | 10.2 | 2.000 | 58.0 | 43 | 7.267 | 5.788 | 2.000 | 23.0 | 15 |
| *External border matching:* | | | | | | | | | | |
| 1970 Population | 34,647 | 63,901 | 6,010 | 253,118 | 14 | 41,022 | 72,711 | 5,879 | 266,815 | 21 |
| 1989 Ind. Employment | 3,771 | 6,317 | 163.0 | 21,598 | 11 | 4,166 | 6,473 | 147.0 | 22,022 | 20 |
| 1989 Ind. Turnover | 98,293 | 130,617 | 2,523 | 382,986 | 11 | 104,737 | 131,277 | 2,080 | 386,449 | 20 |
| 1989 Manufacturing | 51,211 | 86,545 | 0.000 | 281,351 | 11 | 57,188 | 88,302 | 0.000 | 302,282 | 20 |
| 1989 SIC | 8,169 | 4,720 | 2,730 | 20,110 | 9 | 9,170 | 5,908 | 2,730 | 20,110 | 48 |
| Military count | 1.333 | 0.577 | 1.000 | 2.000 | 3 | 1.500 | 0.577 | 1.000 | 2.000 | 4 |
| Military size | 2.667 | 1.155 | 2.000 | 4.000 | 3 | 3.000 | 1.155 | 2.000 | 4.000 | 4 |

*Notes:* Each row reports summary statistics of the matching variable for treated (border) and control (interior) cities in the corresponding matched sample. Each matching exercise is independent: the row for "1970 Population" under internal border matching reports balance in the sample matched on 1970 population (Table 3, Column 1 for Europe; Table A9, Column 1 for Central Asia). Columns (1)–(4) and (6) of the matching tables use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section. The 1:2 ratio balances two considerations: using more than one neighbor improves efficiency in the small Central Asian samples, while restricting to two avoids drawing poor matches from distant regions of the covariate distribution. With replacement and a fixed neighbor count, exact balance on the matching variable is not guaranteed, particularly when the treated and control distributions have different shapes or when a few control cities serve as neighbors for many treated cities. Column (5) uses exact matching on each city's primary SIC code, retaining all border and interior cities that shared the same dominant industrial classification in 1989. Military matching uses two covariates (count and workforce size of defense enterprises closed 1989–1991), so both are reported. Statistics are computed on the cross-section (one observation per city).

**Table A14: Robustness to Matching on Pre-1991 Characteristics (Central Asia)**

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A: Internal Borders** | | | | | | |
| IB × Division | 0.597 | 0.511 | 0.607 | 0.733 | 0.026 | 1.424* |
| | (0.424) | (0.481) | (0.507) | (0.468) | (0.408) | (0.711) |
| Matching on | 1970 Pop. | 1989 Ind. Empl. | 1989 Ind. Turn. | 1989 Manuf. | 1989 SIC | Military |
| Outcome mean (%) | 1.332 | 1.344 | 1.382 | 1.389 | 1.548 | 1.593 |
| Cities | 151 | 131 | 129 | 133 | 106 | 58 |
| Observations | 697 | 603 | 595 | 614 | 483 | 266 |
| Adj. $R^2$ | 0.227 | 0.289 | 0.291 | 0.328 | 0.292 | 0.455 |
| **Panel B: External Borders** | | | | | | |
| EB × Division | 3.007* | 1.086 | 0.736 | 2.699 | −0.533 | 3.819* |
| | (1.696) | (1.040) | (1.113) | (2.139) | (0.651) | (1.450) |
| Matching on | 1970 Pop. | 1989 Ind. Empl. | 1989 Ind. Turn. | 1989 Manuf. | 1989 SIC | Military |
| Outcome mean (%) | 1.525 | 1.553 | 1.515 | 1.486 | 1.746 | 2.294 |
| Cities | 21 | 16 | 23 | 19 | 25 | 5 |
| Observations | 91 | 69 | 102 | 83 | 110 | 22 |
| Adj. $R^2$ | 0.229 | 0.192 | 0.083 | 0.012 | 0.306 | 0.451 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* Each column re-estimates the baseline specification on a matched subsample of treated border cities and their nearest neighbors. Columns (1)–(4) and (6) use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section. Column (5) uses exact matching on each city's primary SIC code, restricting the sample to border and interior cities that shared the same dominant industrial classification in 1989. In Central Asia, matched samples are small, particularly for external borders (Panel B), where the treated sample contains only 11–15 cities depending on covariate availability. The negative EB coefficient in Column (5) of Panel B (−0.533) rests on just 25 cities and likely reflects the composition of the few SIC strata that contain both treated and control cities rather than a substantive finding. Column (6) of Panel B retains just 5 cities, so this estimate should not be over-interpreted. "Military" matches jointly on the count and total workforce of defense enterprises that closed between 1989 and 1991. The dependent variable is annualized city population growth (%). "Division" equals one for post-1991 periods. All specifications include city and period fixed effects. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A15: Matching Design Variations (Europe)

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | Internal Borders | | | External Borders | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| IB × Division | −0.505*** | −0.600*** | −0.392*** | | | |
| | (0.132) | (0.131) | (0.138) | | | |
| EB × Division | | | | −0.802*** | −0.699*** | −0.526* |
| | | | | (0.262) | (0.258) | (0.267) |
| Control radius | None | ≤1,000 km | ≤500 km | None | ≤1,000 km | ≤500 km |
| Median pair dist. (km) | 1,326 | 637 | 347 | — | 636 | 334 |
| Treated cities matched | 290 | 290 | 290 | 117 | 117 | 117 |
| Outcome mean (%) | 0.464 | 0.515 | 0.531 | 0.606 | 0.626 | 0.677 |
| Cities | 626 | 588 | 520 | 260 | 217 | 200 |
| Observations | 3,062 | 2,870 | 2,529 | 1,272 | 1,058 | 973 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* Each column re-estimates the baseline specification on a matched subsample. All columns condition on all six pre-1991 covariates simultaneously: 1970 population, 1989 industrial employment, turnover, and manufacturing output, and the count and workforce of defence enterprises closing between 1989 and 1991. Matching is 1:2 nearest-neighbour with replacement under common support, with the propensity score estimated by probit on the city cross-section of the 1,076 post-Soviet European cities with complete covariate data. Columns (1) and (4) impose no geographic restriction. Columns (2)–(3) and (5)–(6) additionally require each matched control to lie within the stated great-circle distance of its treated city, following Redding and Sturm (2008), who impose a geographic band on their control group; because `psmatch2` offers no caliper on an auxiliary variable, the nearest-neighbour selection in these columns is performed manually on the estimated score. Every treated city retains two matched controls at each radius, except that two external-border cities retain one at 500 km. The median unconstrained pair distance for external borders is not reported because that match is reported in Column (4) only. "Cities" reports unique cities in the matched sample. The dependent variable is annualized city population growth (%). "Division" equals one for post-1991 periods. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Table A16: Excluding Science Cities and Closed Cities

| | Dependent Variable: Population Growth | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| IB × Division | −0.395*** | −0.392*** | −0.379*** | −0.365*** | −0.361*** | 0.479 |
| | (0.110) | (0.110) | (0.111) | (0.112) | (0.113) | (0.384) |
| EB × Division | −0.609*** | −0.607*** | −0.600*** | −0.592*** | −0.593*** | 0.833* |
| | (0.200) | (0.200) | (0.200) | (0.201) | (0.202) | (0.459) |
| Outcome mean (%) | 0.553 | 0.554 | 0.547 | 0.540 | 0.543 | 1.355 |
| Sample | Europe | Europe | Europe | Europe | Europe | CA |
| Science cities excl. | 3 km | 5 km | 10 km | 20 km | 20 km | |
| Closed cities excl. | | | | | ✓ | ✓ |
| Cities | 1,170 | 1,166 | 1,150 | 1,119 | 1,112 | 162 |
| Observations | 5,749 | 5,729 | 5,649 | 5,494 | 5,460 | 745 |
| City FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders, respectively; "Division" equals one for post-1991 periods. Each column re-estimates the baseline specification after excluding cities that hosted Soviet place-based strategic investment, to test whether the post-1991 divergence reflects the collapse of that investment rather than border hardening. Science Cities are matched to sample cities on coordinates, because Schweiger et al. (2022) define them at the level of the Russian municipality, many places were renamed after 1991, and statistics were never published for closed cities; roughly a third were built from scratch and fall below the 1970 population threshold. The exclusion removes 51 sample cities at 3 km and 102 at 20 km, since one Science City can neighbor several sample towns. Only one matched Science City lies within 75 km of an internal border. Columns (5)–(6) additionally exclude 20 closed or restricted-access cities outside Russia, hand-coded from secondary sources. Columns (1)–(4) exclude cities matched to the 95 Science Cities identified by Schweiger et al. (2022) at successively wider radii; the list covers the territory of modern Russia only, so these exclusions bear on the European sample. Matching is by coordinates, with the reported radius applied to the distance between each sample city and the nearest Science City. Only one lies within 75 km of an internal border. Widening the radius beyond 10 km removes adjacent towns rather than additional Science Cities, so Column (4) should be read as a neighbourhood exclusion. Columns (5)–(6) additionally exclude 20 closed or restricted-access cities outside Russia, hand-coded from secondary sources: 7 in post-Soviet Europe (Paldiski, Sillamäe, Liepāja, Sevastopol, Zhovti Vody, Kamianske, and Dnipro) and 13 in Central Asia (Baikonur, Stepnogorsk, Aktau, Ust-Kamenogorsk, Chkalovsk, Taboshar, Khaidarkan, Kadamjay, Kara-Balta, Mailuu-Suu, Zarafshon, Chirchik, and Navoiy). Because closed cities were largely omitted from the published Soviet statistical sources on which the population panel is built, several of the most restricted sites (Sarov, Ozyorsk, Seversk, Novouralsk, Lesnoy, Kurchatov, Sniečkus, and Slavutych) do not appear in the sample at all. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Appendix A1 - Data Cleaning

*Defence Industry Data Matching*

The raw database contains over 33,000 enterprise records spanning the full Soviet period. I restrict the sample to enterprises whose military work ended between 1989 and 1991, the window surrounding dissolution, and whose workforce size is documented (categories 1 through 3 on the database's ordinal scale). Enterprises that ceased military production before 1970 or began after 1991 are excluded. Because city names in the defense database do not map directly to the population panel, I match locations using a combination of automated string matching and manual resolution, successfully linking 822 in the filtered database to cities in the main sample. For each matched city, I construct two variables: the count of defense enterprises that stopped work in the 1989–1991 window, and the sum of their ordinal workforce sizes.

*Night Lights Data*

Several steps were undertaken to transform pixel-level lights data into city-level data following Storeygard (2016). First, 30 satellite-years of night lights data were merged into a single binary grid, indicating whether a pixel was lit above a digital number threshold of 20 in at least one year.[12] These ever-lit areas were then converted into polygons by aggregating contiguous lit pixels. For each polygon and year, I summed the raw digital number values (ranging from 0 to 63) of all constituent pixels to obtain a continuous measure of total light intensity. The regression outcome is the log of this summed intensity, which captures variation in economic activity across cities and over time.

I matched polygons to known cities using census population data with latitude and longitude coordinates from Brinkhoff (2021). Polygons that did not correspond to a known city were excluded; these likely represent forest fires, sensor noise not flagged by the satellite algorithm, or smaller towns and large villages, and account for 10 to 20 percent of total digital numbers in the 15-country sample.[13]

Additionally, I removed lights from gas flares, as identified by Elvidge et al. (2009). To further minimize errors, I clipped the nighttime lights raster to the national boundaries of FSU countries. Without this step, light emitted by cities in neighboring non-FSU countries (e.g., Poland, Finland, China) could bleed into FSU border areas through the well-documented blooming effect in DMSP data, artificially inflating light intensity for FSU cities near external borders. The night lights dataset is annual from 1992 to 2013, enabling analysis only for the period following the dissolution.[14]

[12] The threshold of 20 is applied only at the polygon construction stage to define city boundaries by identifying reliably lit urban areas; it does not affect the outcome variable used in regressions. Li et al. (2020) show that thresholds of 20 to 30 effectively eliminate blooming artifacts and sensor noise around urban peripheries while retaining reliably lit areas.

[13] An alternative approach would use all lit pixels within a fixed buffer around each city centroid, without a brightness threshold. This would avoid potential censoring of newly developed areas at urban peripheries but would also introduce noise from non-urban light sources. The polygon-based approach used here ties the nightlights outcome directly to known urban footprints identified in the census data, ensuring that the measured light corresponds to the same cities whose population growth is analyzed in the main results.

[14] Gibson et al. (2021) highlight key issues with DMSP night lights data, such as blurring and top-coding, which can distort economic measurements. They particularly note that these errors lead to poor predictions of economic activity in low-density rural areas. However, the study focuses on whole cities with populations of at least five thousand people in 1970, where these data provide more reliable insights. Additionally, the results are robust to dropping large cities such as Moscow, where top-coding may be an issue.

## Appendix B - Effects by Individual Border Pair

To further probe how the effect differs for each border pair, I estimate the following three-way interaction model separately for internal and external borders:

$$y_{ct} = \sum_{j=1}^{J} \beta_j \big(\text{border}_c \times \text{division}_t \times \mathbf{1}\{\text{borderpair}_c = j\}\big) + \alpha_c + d_t + \epsilon_{ct}, \tag{7}$$

where $\text{border}_c$ indicates proximity to internal or external borders, $\mathbf{1}\{\text{borderpair}_c = j\}$ is an indicator equal to one if city $c$'s nearest border belongs to border pair $j$ and zero otherwise, and $J$ is the total number of border pairs. City fixed effects $\alpha_c$ absorb all time-invariant city characteristics, including border-pair membership, while year fixed effects $d_t$ control for common census year shocks. This model allows isolation of how the Soviet Union's dissolution uniquely influenced population growth around distinct borders.

Figure B1 presents results for internal border pairs. In Europe, the majority of border pairs exhibit negative coefficients, though magnitudes and precision vary considerably with sample size. The largest and most precisely estimated declines are at Latvia–Lithuania ($-1.70$, significant at the 1% level, 31 cities), Moldova–Ukraine ($-1.61$, significant at the 1% level, 46 cities), Belarus–Lithuania ($-1.43$, significant at the 1% level, 22 cities), and Belarus–Ukraine ($-1.07$, significant at the 1% level, 26 cities). Armenia–Azerbaijan shows a large negative effect ($-1.76$, significant at the 1% level), though the Nagorno-Karabakh conflict complicates interpretation. Belarus–Russia ($-0.35$, significant at the 5% level) and Lithuania–Russia ($-1.09$, significant at the 1% level) are also negative, while Russia–Ukraine is modestly positive (0.30, significant at the 5% level), potentially reflecting the continued dominance of bilateral Russia–Ukraine trade despite border frictions. This uplift requires specific context: rather than cleanly severing ties, the economies of eastern Ukraine and western Russia remained deeply intertwined throughout the 1990s, particularly in heavy industry, metallurgy, and energy transit. The imposition of a national boundary transformed these previously frictionless neighboring cities into essential customs, logistics, and transit hubs for the largest bilateral trade corridor in the former Soviet Union. Several pairs with very few cities (Belarus–Latvia with 1 city, Azerbaijan–Russia with 7 cities, Latvia–Russia with 4 cities) produce extreme or imprecise estimates that should not be over-interpreted.

In Central Asia, six of the seven internal border pairs show positive coefficients, with Kazakhstan–Russia the exception. Tajikistan–Uzbekistan (1.05, $p = 0.09$) and Turkmenistan–Uzbekistan (1.08, $p = 0.10$) are the most precisely estimated, though the intervals remain wide throughout. None of the seven excludes zero at conventional levels, so the pair-level evidence does not overturn the aggregate Central Asian internal border null in Table 1, and the uniformly small magnitudes are consistent with the limited disruption expected from enforcing borders across weakly integrated economies.

Figure B2 presents results for external border pairs. In Europe, the largest and most precisely estimated negative effects are at Moldova–Romania ($-1.78$, significant at the 1% level, 26 cities) and Lithuania–Poland ($-1.97$, significant at the 1% level, 9 cities), both on the western frontier; all nine Lithuanian towns also lie within 75 km of an internal frontier, so their decline compounds the two shocks (Table A4). Belarus–Poland is also negative ($-0.58$, significant at the 5% level). Several Caucasian border pairs show mixed results: Armenia–Turkey ($-0.58$) and Georgia–Turkey ($-0.35$) are negative but imprecise, consistent with the limited commercial potential of these frontiers noted in Section 2. Poland–Russia shows a notable positive effect (1.01, significant at the 1% level), likely reflecting Kaliningrad's unique economic arrangements as a Russian exclave with special trade provisions. Border pairs with very few cities (Hungary–Ukraine with 2 cities, Finland–Russia with 4 cities) are imprecise.

In Central Asia, all external border pairs show positive coefficients. Iran–Turkmenistan has the largest effect (3.68, significant at the 1% level), followed by Afghanistan–Tajikistan (1.55, significant at the 5% level). China–Kazakhstan (0.68, $p = 0.13$) is positive but imprecise, reflecting the single city (Zharkent) near this border. The uniformly positive direction across geographically distinct frontiers reinforces the interpretation that border opening generated market access gains in Central Asia. To reiterate, individual estimates are imprecise for pairs with few cities, so the value of this exercise lies in the broad pattern rather than any single coefficient.

Figure B2 also reports Russia's Asian frontiers, shown separately under "Other" because they face different neighbours from either regional group. These cities are external border cities on the same 75 km rule as all others and are estimated within the European regression, since Russia is classified with post-Soviet Europe throughout (Figure 1). The China–Russia estimate is $-1.524$ on 11 cities ($p = 0.23$) and Mongolia–Russia is 0.847 on 2 cities ($p = 0.15$). Neither is distinguishable from zero and the two carry opposite signs, so the pair-level evidence does not identify an effect on either frontier. The China–Russia estimate does not provide evidence of a comparable positive effect, but does not establish a difference from Central Asia. The Central Asian estimate pools frontiers with Afghanistan and Iran alongside China, and the China–Kazakhstan cell contains a single city, so the regional gain cannot be attributed specifically to Chinese market access.

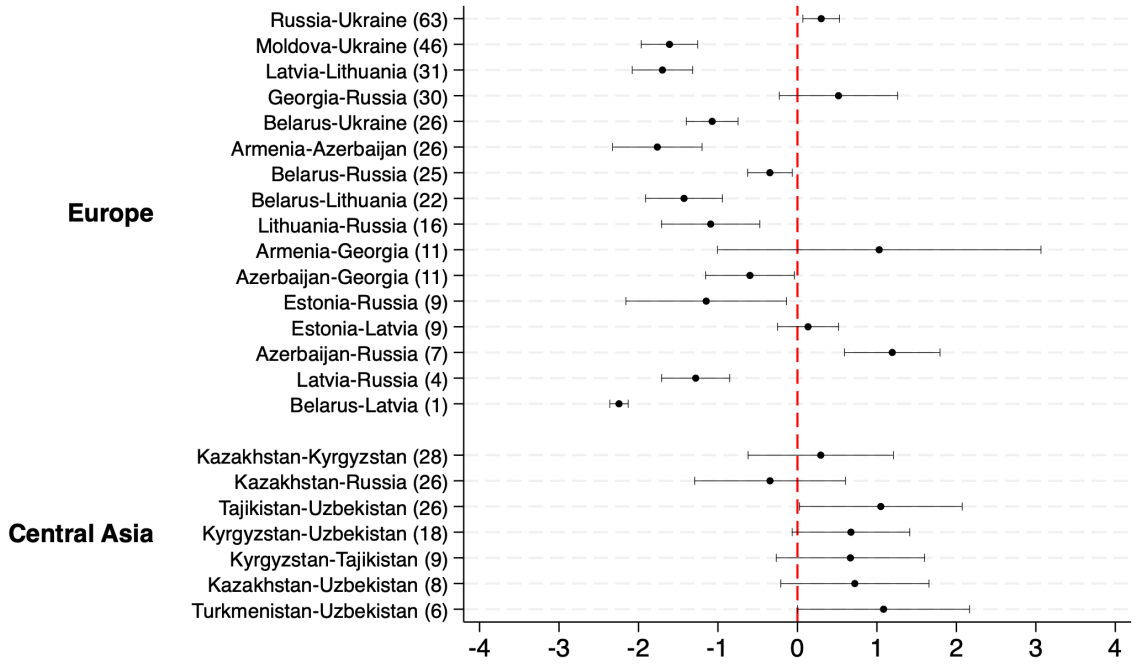



Figure B1: Treatment Effects by Internal Border Pair

*Notes: Each coefficient represents the post-1991 effect on annualized population growth for cities within 75 km of the indicated internal border pair, estimated from equation 7. Separate regressions are run for Europe and Central Asia, with interior cities in each region as controls. Numbers in parentheses indicate the number of border cities per pair. Border pairs sorted by city count within each region. All specifications include city, period, and border-pair fixed effects. Standard errors clustered at the city level. Bars show 90% confidence intervals.*

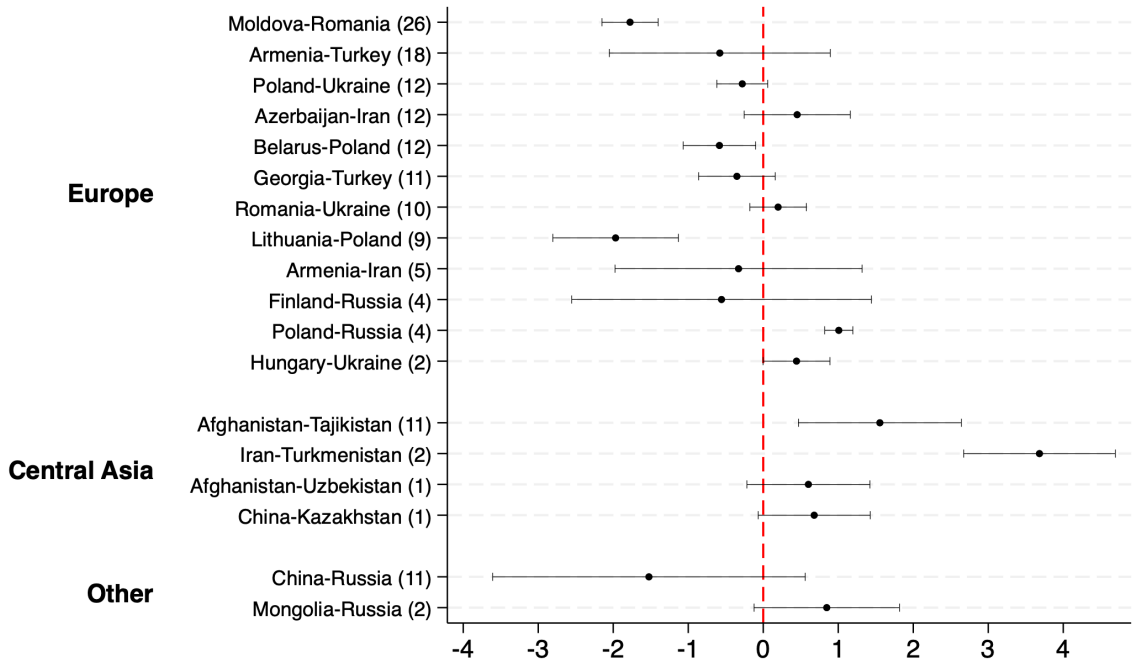



Figure B2: Treatment Effects by External Border Pair

*Notes:* Each coefficient represents the post-1991 effect on annualized population growth for cities within 75 km of the indicated external border pair, estimated from equation 7. Two regressions are run, one for post-Soviet Europe and one for Central Asia, with interior cities in each region as controls. The groupings are for display only: pairs labelled "Other" are Russia's Asian frontiers and are estimated within the European regression, since Russia is classified with post-Soviet Europe throughout (Figure 1). Numbers in parentheses indicate the number of treated border cities per pair.Two pairs are estimated but omitted from the plot: Slovakia–Ukraine, which contains one city and is collinear, and Estonia–Finland, where the three cities reach Finland across the Gulf of Finland rather than over a land frontier. All specifications include city, period, and border-pair fixed effects. Standard errors clustered at the city level. Bars show 90% confidence intervals.

## Appendix C - Effects on Economic Activity Proxied by Night Lights

In this section, I examine whether the city-level population results for Europe and Central Asia also manifest in nighttime lights data, a widely used proxy for local economic activity. Although nighttime light data is only available from 1992 onward, I exploit this limitation by treating 1992, the first full year after the Soviet dissolution, as the omitted baseline. I estimate separate event-study specifications for internal and external border cities, comparing changes in log night lights relative to interior cities over the period 1993–2013.

$$\log(\text{lights})_{ct} = \sum_{\tau=1993}^{2013} \beta_\tau \big(\text{border}_c \times \mathbf{1}\{t = \tau\}\big) + \alpha_c + d_t + \epsilon_{ct}, \tag{8}$$

where $\log(\text{lights})_{ct}$ is the log of summed lights in city $c$ in year $t$, $\text{border}_c$ is a dummy equal to one if a city is within 75 km of a border and zero otherwise, and $\mathbf{1}\{t = \tau\}$ is an indicator for year $\tau$. The year 1992 serves as the omitted baseline, so each coefficient $\beta_\tau$ captures the difference in log night lights between border and interior cities in year $\tau$ relative to 1992. City fixed effects $\alpha_c$ absorb time-invariant city characteristics, while year fixed effects $d_t$ control for common annual shocks. The specification is estimated separately for internal border cities (IB) and external border cities (EB), with the two sets of coefficients plotted together in Figure C1. While this approach does not allow for direct observation of pre-1992 trends, it reveals important information about the relative trajectories of economic activity following the dissolution.

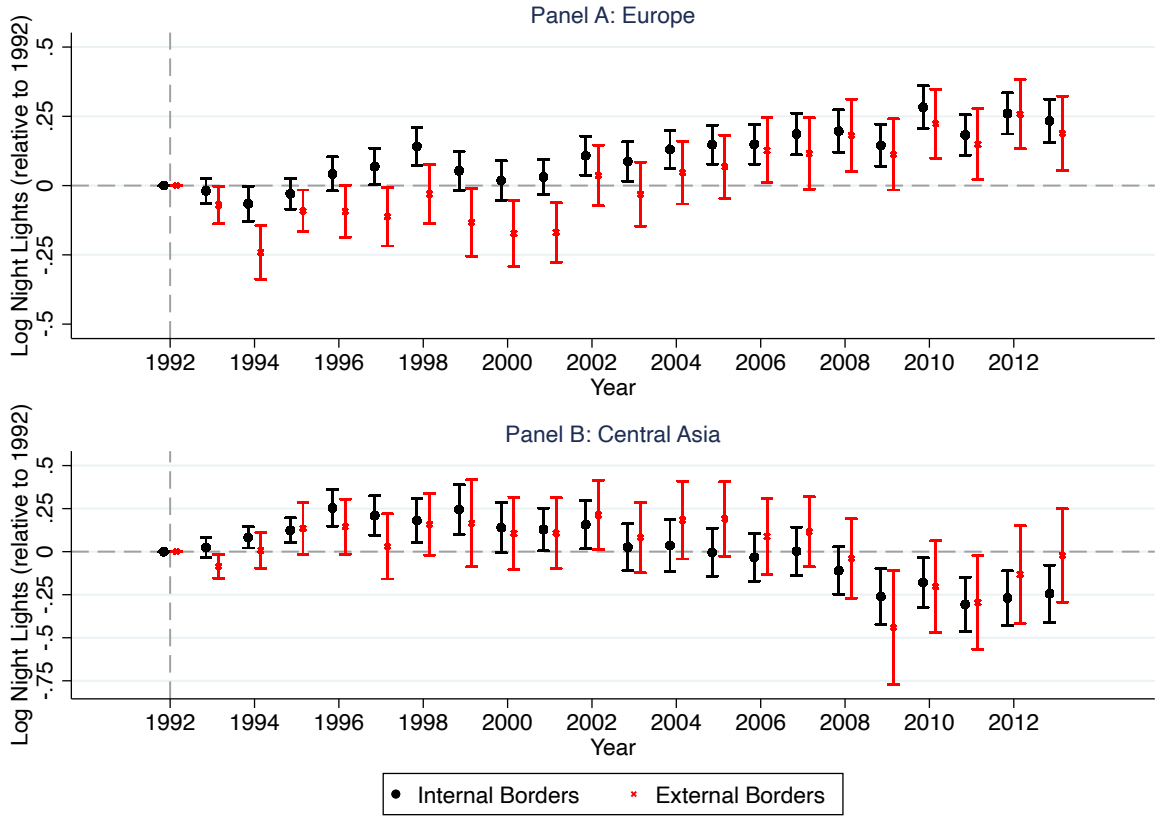



Figure C1: Event-study Estimates of Nighttime Light Intensity relative to 1992

*Notes: This figure plots coefficients from event-study regressions of log nighttime light intensity on border proximity interacted with year indicators, relative to the 1992 baseline. Black circles denote internal borders; red X's denote external borders. Panel A shows Europe; Panel B shows Central Asia. Because nightlights data begin in 1992, no pre-treatment trends can be assessed. All regressions include city, year, and border-pair fixed effects. Standard errors clustered at the city level; bars show 90% confidence intervals.*

Figure C1 presents results for Europe (Panel A) and Central Asia (Panel B). Two important caveats apply when interpreting these results alongside the main population findings. First, nighttime lights data begin in 1992, so there is no pre-treatment period to validate parallel trends. Second, the population regressions estimate annualized growth rates (a flow measure), while the nightlights specification estimates log levels relative to a 1992 baseline. A negative coefficient in the population regression implies continuously diverging population levels, whereas the nightlights coefficients capture the cumulative deviation in economic activity relative to 1992. These differences in outcome measurement mean the two sets of results need not produce identical time-series patterns.

In Europe (Panel A), internal border cities show an initial relative decline in 1994, followed by a rebound that produces positive deviations from the mid-1990s onward. External border cities follow a similar but delayed trajectory, with coefficients near zero through the early 2000s before turning consistently positive after 2005. At first glance, these positive deviations appear to contradict the population results, which show sustained negative effects. However, the two findings are not necessarily inconsistent. The population results capture relative *growth rates*: border cities grew more slowly than interior cities in every post-1991 period. The nightlights results capture relative *levels* of economic activity measured from a 1992 baseline that is itself already depressed by the initial collapse. The positive nightlights deviations after 2002 may reflect catch-up in economic activity per capita as border city populations shrank while economic output partially stabilized, or they may reflect the normalization of border institutions, EU accession of neighboring countries, and increased cross-border cooperation. The nightlights evidence is therefore best interpreted as showing that the acute economic disruption at borders was concentrated in the 1990s, with some recovery in relative economic activity thereafter, even as population continued to decline in relative terms.

In Central Asia (Panel B), internal border cities show positive and increasing deviations through the late 1990s, consistent with initially limited disruption from the enforcement of inter-republican borders. However, these positive trajectories flatten and turn negative from the mid-2000s onward, potentially reflecting increasing state control, higher tariffs, and tighter border enforcement in countries such as Uzbekistan and Turkmenistan. External border cities present a pattern that contrasts sharply with their European counterparts. Whereas European external border cities showed near-zero or negative deviations through the early 2000s, Central Asian external border cities display positive coefficients from the mid-1990s through the early 2000s, with the 2002 coefficient marginally significant. This divergence is consistent with the core finding that the same geopolitical rupture generated opposite market access shocks across regions: European external border cities faced contracting, westward-reorienting economies across the frontier while Central Asian external border cities gained access to previously sealed frontiers. The Central Asian external border gains attenuate and turn negative from 2009 onward, coinciding with the formalization of border regimes and reduced scope for informal shuttle trade. The imprecision of many individual-year coefficients reflects the small sample of external border cities (57 clusters) and the inherent noisiness of DMSP data in sparsely populated regions.

Overall, the nightlights evidence provides partial corroboration of the population findings. The cross-regional contrast in external border trajectories, negative in Europe and positive in Central Asia during the first post-collapse decade, mirrors the opposing signs documented in the population regressions and reinforces the market access interpretation. The two outcome measures capture different dimensions of the border shock, however, and should not be expected to produce identical time-series patterns.